

QuestaSim-64 vcover 2021.1 Coverage Utility 2021.01 Jan 19 2021  
 Start time: 17:18:13 on May 21,2026  
 vcover report -details build/merged.ucdb  
 Coverage Report by instance with details

```

=====
=== Instance: /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva
=== Design Unit: work.spi_regfile_sva
=====
  
```

Assertion Coverage:

| Assertions                                                                |                            | 18          | 18         | 0       | 100.00% |       |
|---------------------------------------------------------------------------|----------------------------|-------------|------------|---------|---------|-------|
| -----                                                                     |                            |             |            |         |         |       |
| Name                                                                      | File(Line)                 | Failure     | Pass       | Vacuous |         |       |
| Disable                                                                   | Attempt                    | Active Peak | Active ATV | Count   | Count   | Count |
| Count                                                                     | Count                      | Count       | Count      |         |         |       |
| -----                                                                     |                            |             |            |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_1_SETUP_THEN_ACCESS      |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(32)  | 0           | 3129374    | 6270705 |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_2_PENABLE_NEEDS_PSEL     |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(40)  | 0           | 3129375    | 6270704 |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_3a_ADDR_CTRL_STABLE      |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(48)  | 0           | 3129374    | 6270705 |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_3b_WDATA_STABLE_ON_WRITE |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(56)  | 0           | 2119       | 9397960 |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_4a_PRDATA_RESERVED_ZERO  |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(65)  | 0           | 5          | 9400074 |         |       |
| 70                                                                        | 9400149                    | 0           | 1 off      |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_4b_PRDATA_WO_REG_ZERO    |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(74)  | 0           | 4          | 9400075 |         |       |
| 70                                                                        | 9400149                    | 0           | 1 off      |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_5_PREADY_ALWAYS_1        |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(82)  | 0           | 3129375    | 6270704 |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_6_PSLVERR_ALWAYS_0       |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(90)  | 0           | 9400079    | 0       |         |       |
| 70                                                                        | 9400149                    | 0           | 10 off     |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_1_TX_OVF_SET_ON_OVERFLOW |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(98)  | 0           | 3          | 9400076 |         |       |
| 70                                                                        | 9400149                    | 0           | 2 off      |         |         |       |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_2_IRQ_EQUATION           |                            |             |            |         |         |       |
|                                                                           | assertions/spi_sva.sv(106) | 0           | 9400079    | 0       |         |       |

|                                                                              |         |   |        |   |         |         |
|------------------------------------------------------------------------------|---------|---|--------|---|---------|---------|
| 70                                                                           | 9400149 | 0 | 10 off |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3a_TX_EMPTY_STICKY          |         |   |        |   |         |         |
| assertions/spi_sva.sv(115)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 9395463 | 4614    |
| 70                                                                           | 9400149 | 2 | 18 off |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3b_RX_FULL_STICKY           |         |   |        |   |         |         |
| assertions/spi_sva.sv(124)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 1410    | 9398669 |
| 70                                                                           | 9400149 | 0 | 4 off  |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3c_TX_OVF_STICKY            |         |   |        |   |         |         |
| assertions/spi_sva.sv(133)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 225     | 9399854 |
| 70                                                                           | 9400149 | 0 | 4 off  |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3d_RX_OVF_STICKY            |         |   |        |   |         |         |
| assertions/spi_sva.sv(142)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 1562    | 9398517 |
| 70                                                                           | 9400149 | 0 | 2 off  |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3e_DONE_STICKY              |         |   |        |   |         |         |
| assertions/spi_sva.sv(151)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 8859498 | 540579  |
| 70                                                                           | 9400149 | 2 | 18 off |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_4_INT_STAT_ZERO_AFTER_RESET |         |   |        |   |         |         |
| assertions/spi_sva.sv(159)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 10      | 9400069 |
| 70                                                                           | 9400149 | 0 | 10 off |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/FIFO_2_RX_COUNT_BOUNDED         |         |   |        |   |         |         |
| assertions/spi_sva.sv(167)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 9400079 | 0       |
| 70                                                                           | 9400149 | 0 | 10 off |   |         |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/FIFO_3_RX_OVF_SET_ON_OVERFLOW   |         |   |        |   |         |         |
| assertions/spi_sva.sv(175)                                                   |         |   |        |   |         |         |
|                                                                              |         |   |        | 0 | 24      | 9400055 |
| 70                                                                           | 9400149 | 0 | 1 off  |   |         |         |

=====  
 === Instance: /tb\_top/u\_wrap/u\_dut/u\_regfile

=== Design Unit: work.apb\_regfile  
 =====

#### Branch Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| -----            | ---- | ---- | -----  | -----    |
| Branches         | 55   | 55   | 0      | 100.00%  |

=====Branch Details=====

Branch Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_regfile

| Line                             | Item | Count | Source |
|----------------------------------|------|-------|--------|
| ----                             | ---- | ----- | -----  |
| File ./golden_rtl/apb_regfile.sv |      |       |        |

```

-----IF Branch-----
168                                6261184    Count coming in to IF
168                                3128394
                                3132790    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----CASE Branch-----
169                                3128394    Count coming in to CASE
170                                26
171                                3127052
172                                4
173                                1194
177                                8
178                                7
179                                19
180                                72
181                                6
182                                6
Branch totals: 10 hits of 10 branches = 100.00%

```

```

-----IF Branch-----
174                                1194    Count coming in to IF
174                                207
174                                987
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
196                                7037    Count coming in to IF
196                                625
                                6412    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----CASE Branch-----
198                                625    Count coming in to CASE
199                                601
200                                11
201                                13
Branch totals: 3 hits of 3 branches = 100.00%

```

```

-----IF Branch-----
213                                9351    Count coming in to IF
213                                30
226                                9321
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
227                                9321    Count coming in to IF
227                                2120
                                7201    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----CASE Branch-----
228                                2120    Count coming in to CASE
229                                389
237                                204
238                                451
242                                82
243                                170
244                                824
Branch totals: 6 hits of 6 branches = 100.00%

```

```

-----IF Branch-----
256                                9321    Count coming in to IF
256                                192
                                9129    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
260                                9321    Count coming in to IF
260                                3
                                9318    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
262                                9321    Count coming in to IF
262                                24
                                9297    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
264                                9321    Count coming in to IF
264                                9
                                9312    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
266                                9321    Count coming in to IF
266                                378
                                8943    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
268                                9321    Count coming in to IF
268                                621
                                8700    All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
284                                2941    Count coming in to IF
284                                20

```

```

288          1          380
291          1          2541
Branch totals: 3 hits of 3 branches = 100.00%

```

```

-----IF Branch-----
292          2541      Count coming in to IF
292          1          622
1919      All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
296          2541      Count coming in to IF
296          1          621
1920      All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
306          3022      Count coming in to IF
306          1          20
310          1          380
313          1          2622
Branch totals: 3 hits of 3 branches = 100.00%

```

```

-----IF Branch-----
314          2622      Count coming in to IF
314          1          597
2025      All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

-----IF Branch-----
318          2622      Count coming in to IF
318          1          593
2029      All False Count
Branch totals: 2 hits of 2 branches = 100.00%

```

```

Condition Coverage:
Enabled Coverage      Bins    Covered    Misses    Coverage
-----
Conditions            14       13         1     92.85%

```

=====Condition Details=====

Condition Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_regfile --

File ./golden\_rtl/apb\_regfile.sv

-----Focused Condition View-----

```

Line      196 Item      1 ((apb_write && (PADDR == 8)) && ctrl_en)
Condition totals: 2 of 3 input terms covered = 66.66%

```

| Input Term   | Covered | Reason for no coverage | Hint     |
|--------------|---------|------------------------|----------|
| apb_write    | Y       |                        |          |
| (PADDR == 8) | Y       |                        |          |
| ctrl_en      | N       | '_0' not hit           | Hit '_0' |

| Rows:  | Hits    | FEC Target     | Non-masking condition(s)    |
|--------|---------|----------------|-----------------------------|
| Row 1: | 10      | apb_write_0    | -                           |
| Row 2: | 10      | apb_write_1    | (ctrl_en && (PADDR == 8))   |
| Row 3: | 10      | (PADDR == 8)_0 | apb_write                   |
| Row 4: | 10      | (PADDR == 8)_1 | (ctrl_en && apb_write)      |
| Row 5: | ***0*** | ctrl_en_0      | (apb_write && (PADDR == 8)) |
| Row 6: | 10      | ctrl_en_1      | (apb_write && (PADDR == 8)) |

-----Focused Condition View-----

Line 256 Item 1 (apb\_write && (PADDR == 28))  
Condition totals: 2 of 2 input terms covered = 100.00%

-----Focused Condition View-----

Line 262 Item 1 (rx\_push\_valid && rx\_full\_w)  
Condition totals: 2 of 2 input terms covered = 100.00%

-----Focused Condition View-----

Line 264 Item 1 ((rx\_push\_valid && ~rx\_full\_w) && (rx\_count == (8 - 1)))  
Condition totals: 3 of 3 input terms covered = 100.00%

-----Focused Condition View-----

Line 266 Item 1 (tx\_pop && (tx\_count == 1))  
Condition totals: 2 of 2 input terms covered = 100.00%

-----Focused Condition View-----

Line 314 Item 1 (rx\_push\_valid && ~rx\_full\_w)  
Condition totals: 2 of 2 input terms covered = 100.00%

Expression Coverage:

| Enabled Coverage | Bins | Covered | Misses | Coverage |
|------------------|------|---------|--------|----------|
| -----            | ---- | ----    | -----  | -----    |
| Expressions      | 14   | 13      | 1      | 92.85%   |

=====Expression Details=====

Expression Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_regfile --

File ./golden\_rtl/apb\_regfile.sv

-----Focused Expression View-----

Line 107 Item 1 (PSEL & PENABLE)  
Expression totals: 1 of 2 input terms covered = 50.00%

| Input Term | Covered | Reason for no coverage | Hint     |
|------------|---------|------------------------|----------|
| PSEL       | N       | '_0' not hit           | Hit '_0' |
| PENABLE    | Y       |                        |          |

| Rows:  | Hits    | FEC Target | Non-masking condition(s) |
|--------|---------|------------|--------------------------|
| Row 1: | ***0*** | PSEL_0     | PENABLE                  |
| Row 2: | 10      | PSEL_1     | PENABLE                  |
| Row 3: | 10      | PENABLE_0  | PSEL                     |
| Row 4: | 10      | PENABLE_1  | PSEL                     |

-----Focused Expression View-----

Line 108 Item 1 (apb\_access & PWRITE)  
Expression totals: 2 of 2 input terms covered = 100.00%

-----Focused Expression View-----

Line 109 Item 1 (apb\_access & ~PWRITE)  
Expression totals: 2 of 2 input terms covered = 100.00%

-----Focused Expression View-----

Line 122 Item 1 (tx\_count == 8)  
Expression totals: 1 of 1 input term covered = 100.00%

-----Focused Expression View-----

Line 123 Item 1 (tx\_count == 0)  
Expression totals: 1 of 1 input term covered = 100.00%

-----Focused Expression View-----

Line 128 Item 1 (rx\_count == 8)  
Expression totals: 1 of 1 input term covered = 100.00%

-----Focused Expression View-----

Line 129 Item 1 (rx\_count == 0)  
Expression totals: 1 of 1 input term covered = 100.00%

-----Focused Expression View-----

Line 206 Item 1 (tx\_push\_valid & ~tx\_full\_w)  
Expression totals: 2 of 2 input terms covered = 100.00%

-----Focused Expression View-----

Line 207 Item 1 (tx\_push\_valid & tx\_full\_w)  
Expression totals: 2 of 2 input terms covered = 100.00%

Statement Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| Statements       | 96   | 96   | 0      | 100.00%  |

=====Statement Details=====

Statement Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_regfile --

| Line                             | Item | Count   | Source |
|----------------------------------|------|---------|--------|
| ----                             | ---- | -----   | -----  |
| File ./golden_rtl/apb_regfile.sv |      |         |        |
| 107                              | 1    | 9388135 |        |
| 108                              | 1    | 6260880 |        |
| 109                              | 1    | 6260880 |        |
| 121                              | 1    | 1447    |        |
| 122                              | 1    | 1263    |        |
| 123                              | 1    | 1263    |        |
| 127                              | 1    | 1394    |        |
| 128                              | 1    | 1211    |        |
| 129                              | 1    | 1211    |        |
| 132                              | 1    | 1331    |        |
| 137                              | 1    | 2776    |        |
| 148                              | 1    | 408     |        |
| 158                              | 1    | 460     |        |
| 159                              | 1    | 46      |        |
| 160                              | 1    | 563     |        |
| 161                              | 1    | 174     |        |
| 162                              | 1    | 143     |        |
| 165                              | 1    | 6261184 |        |
| 166                              | 1    | 6261184 |        |
| 167                              | 1    | 6261184 |        |
| 170                              | 1    | 26      |        |
| 171                              | 1    | 3127052 |        |
| 172                              | 1    | 4       |        |
| 174                              | 1    | 1194    |        |
| 175                              | 1    | 1194    |        |
| 177                              | 1    | 8       |        |
| 178                              | 1    | 7       |        |
| 179                              | 1    | 19      |        |
| 180                              | 1    | 72      |        |
| 181                              | 1    | 6       |        |
| 182                              | 1    | 6       |        |
| 193                              | 1    | 7037    |        |
| 194                              | 1    | 7037    |        |
| 195                              | 1    | 7037    |        |
| 197                              | 1    | 625     |        |
| 199                              | 1    | 601     |        |
| 200                              | 1    | 11      |        |
| 201                              | 1    | 13      |        |
| 206                              | 1    | 1286    |        |
| 207                              | 1    | 1286    |        |
| 212                              | 1    | 9351    |        |
| 214                              | 1    | 30      |        |
| 215                              | 1    | 30      |        |



|     |   |      |
|-----|---|------|
| 216 | 1 | 30   |
| 217 | 1 | 30   |
| 218 | 1 | 30   |
| 219 | 1 | 30   |
| 220 | 1 | 30   |
| 221 | 1 | 30   |
| 222 | 1 | 30   |
| 223 | 1 | 30   |
| 224 | 1 | 30   |
| 225 | 1 | 30   |
| 230 | 1 | 389  |
| 231 | 1 | 389  |
| 232 | 1 | 389  |
| 233 | 1 | 389  |
| 234 | 1 | 389  |
| 235 | 1 | 389  |
| 237 | 1 | 204  |
| 239 | 1 | 451  |
| 240 | 1 | 451  |
| 242 | 1 | 82   |
| 243 | 1 | 170  |
| 254 | 1 | 9321 |
| 257 | 1 | 192  |
| 261 | 1 | 3    |
| 263 | 1 | 24   |
| 265 | 1 | 9    |
| 267 | 1 | 378  |
| 269 | 1 | 621  |
| 271 | 1 | 9321 |
| 276 | 1 | 589  |
| 277 | 1 | 460  |
| 283 | 1 | 2941 |
| 285 | 1 | 20   |
| 286 | 1 | 20   |
| 287 | 1 | 20   |
| 287 | 2 | 160  |
| 287 | 3 | 160  |
| 289 | 1 | 380  |
| 290 | 1 | 380  |
| 293 | 1 | 622  |
| 294 | 1 | 622  |
| 297 | 1 | 621  |
| 305 | 1 | 3022 |
| 307 | 1 | 20   |
| 308 | 1 | 20   |
| 309 | 1 | 20   |
| 309 | 2 | 160  |
| 309 | 3 | 160  |
| 311 | 1 | 380  |
| 312 | 1 | 380  |

|     |   |     |
|-----|---|-----|
| 315 | 1 | 597 |
| 316 | 1 | 597 |
| 319 | 1 | 593 |

Toggle Coverage:

|                  |      |      |        |          |
|------------------|------|------|--------|----------|
| Enabled Coverage | Bins | Hits | Misses | Coverage |
| -----            | ---- | ---- | -----  | -----    |
| Toggles          | 1116 | 673  | 443    | 60.30%   |

=====Toggle Details=====

Toggle Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_regfile --

| "Coverage" | Node                | 1H->0L | 0L->1H |
|------------|---------------------|--------|--------|
| -----      |                     |        |        |
|            | PRDATA[22]          | 0      | 0      |
| 0.00       |                     |        |        |
|            | PRDATA[24]          | 0      | 0      |
| 0.00       |                     |        |        |
|            | PREADY              | 0      | 0      |
| 0.00       |                     |        |        |
|            | PRESETn             | 0      | 10     |
| 50.00      |                     |        |        |
|            | PSLVERR             | 0      | 0      |
| 0.00       |                     |        |        |
|            | clk_div_word[16-31] | 0      | 0      |
| 0.00       |                     |        |        |
|            | ctrl_word[8-31]     | 0      | 0      |
| 0.00       |                     |        |        |
|            | delay_word[8-31]    | 0      | 0      |
| 0.00       |                     |        |        |
|            | i[0-31]             | 0      | 0      |
| 0.00       |                     |        |        |
|            | int_en_word[5-31]   | 0      | 0      |
| 0.00       |                     |        |        |
|            | int_stat_word[5-31] | 0      | 0      |
| 0.00       |                     |        |        |
|            | rx_push_data[20]    | 0      | 2      |
| 50.00      |                     |        |        |
|            | rx_push_data[22]    | 0      | 0      |
| 0.00       |                     |        |        |
|            | rx_push_data[24]    | 0      | 1      |
| 50.00      |                     |        |        |
|            | ss_ctrl_word[4-31]  | 0      | 0      |
| 0.00       |                     |        |        |
|            | ss_val[0-3]         | 0      | 0      |
| 0.00       |                     |        |        |
|            | status_word[7-31]   | 0      | 0      |
| 0.00       |                     |        |        |

|      |                  |   |   |
|------|------------------|---|---|
| 0.00 | tx_push_data[17] | 0 | 0 |
| 0.00 | tx_push_data[22] | 0 | 0 |
| 0.00 | tx_push_data[24] | 0 | 0 |
| 0.00 | tx_push_data[29] | 0 | 0 |
| 0.00 | tx_word[17]      | 0 | 0 |
| 0.00 | tx_word[22]      | 0 | 0 |
| 0.00 | tx_word[24]      | 0 | 0 |
| 0.00 | tx_word[29]      | 0 | 0 |

Total Node Count = 558  
 Toggled Node Count = 335  
 Untoggled Node Count = 223

Toggle Coverage = 60.30% (673 of 1116 bins)

=====  
 === Instance: /tb\_top/u\_wrap/u\_dut/u\_core/u\_core\_sva  
 === Design Unit: work.spi\_core\_sva  
 =====

#### Assertion Coverage:

|            |   |   |   |         |
|------------|---|---|---|---------|
| Assertions | 6 | 6 | 0 | 100.00% |
|------------|---|---|---|---------|

| Name    | File(Line) | Failure     | Pass       | Vacuous |
|---------|------------|-------------|------------|---------|
| Disable | Attempt    | Active Peak | Active ATV |         |
| Count   | Count      | Count       | Count      | Count   |

|                                                                          |         |   |        |                |
|--------------------------------------------------------------------------|---------|---|--------|----------------|
| -----                                                                    |         |   |        |                |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_1_SCLK_IDLE_POLARITY          |         |   |        |                |
| assertions/spi_sva.sv(202)                                               |         |   |        |                |
| 70                                                                       | 9400149 | 0 | 10 off | 0 8327 9391752 |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_2A_MOSI_STABLE_RISING_SAMPLE  |         |   |        |                |
| assertions/spi_sva.sv(210)                                               |         |   |        |                |
| 70                                                                       | 9400149 | 0 | 9 off  | 0 5144 9394935 |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_2B_MOSI_STABLE_FALLING_SAMPLE |         |   |        |                |
| assertions/spi_sva.sv(218)                                               |         |   |        |                |
| 70                                                                       | 9400149 | 0 | 1 off  | 0 224 9399855  |

```

/tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_3_START_REQUIRES_SS_ASSERTED
assertions/spi_sva.sv(226)
0 621 9399458
70 9400149 0 9 off
/tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_5_TX_POP_ONE_CYCLE_PULSE
assertions/spi_sva.sv(234)
0 621 9399458
70 9400149 0 9 off
/tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_6_DONE_PULSE_ONE_CYCLE
assertions/spi_sva.sv(242)
0 621 9399458
70 9400149 0 9 off

```

```

====
=== Instance: /tb_top/u_wrap/u_dut/u_core
=== Design Unit: work.spi_core
=====

```

#### Branch Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| -----            | ---- | ---- | -----  | -----    |
| Branches         | 57   | 55   | 2      | 96.49%   |

#### ====Branch Details=====

Branch Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core

| Line                                          | Item | Count | Source                |
|-----------------------------------------------|------|-------|-----------------------|
| ----                                          | ---- | ----- | -----                 |
| File ./golden_rtl/spi_core.sv                 |      |       |                       |
| -----IF Branch-----                           |      |       |                       |
| 80                                            |      | 27    | Count coming in to IF |
| 80                                            | 1    | 13    |                       |
| 81                                            | 1    | 14    |                       |
| Branch totals: 2 hits of 2 branches = 100.00% |      |       |                       |
| -----IF Branch-----                           |      |       |                       |
| 81                                            |      | 14    | Count coming in to IF |
| 81                                            | 2    | 6     |                       |
| 81                                            | 3    | 8     |                       |
| Branch totals: 2 hits of 2 branches = 100.00% |      |       |                       |
| -----IF Branch-----                           |      |       |                       |
| 89                                            |      | 3936  | Count coming in to IF |
| 89                                            | 1    | 3706  |                       |
| 89                                            | 2    | 230   |                       |
| Branch totals: 2 hits of 2 branches = 100.00% |      |       |                       |
| -----IF Branch-----                           |      |       |                       |
| 96                                            |      | 4771  | Count coming in to IF |
| 96                                            | 1    | 224   |                       |

98 1 4547  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
104 621 Count coming in to IF  
104 1 13  
105 1 608  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
112 9393529 Count coming in to IF  
112 1 30  
130 1 9393499  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
135 9393499 Count coming in to IF  
135 1 387  
142 1 9393112  
Branch totals: 2 hits of 2 branches = 100.00%

-----CASE Branch-----  
143 9393112 Count coming in to CASE  
145 1 2006  
178 1 1575061  
224 1 104819  
246 1 7711226  
259 1 \*\*\*0\*\*\*  
Branch totals: 4 hits of 5 branches = 80.00%

-----IF Branch-----  
149 2006 Count coming in to IF  
149 1 621  
1385 All False Count  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
159 621 Count coming in to IF  
159 1 597  
160 1 24  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
160 24 Count coming in to IF  
160 2 11  
160 3 13  
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----  
165 621 Count coming in to IF

|     |   |     |                 |
|-----|---|-----|-----------------|
| 165 | 1 | 609 |                 |
|     |   | 12  | All False Count |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |     |                       |
|-----|---|-----|-----------------------|
| 166 |   | 609 | Count coming in to IF |
| 166 | 1 | 6   |                       |
| 167 | 1 | 603 |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |     |                       |
|-----|---|-----|-----------------------|
| 168 |   | 603 | Count coming in to IF |
| 168 | 1 | 591 |                       |
| 169 | 1 | 12  |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |    |                       |
|-----|---|----|-----------------------|
| 169 |   | 12 | Count coming in to IF |
| 169 | 2 | 5  |                       |
| 169 | 3 | 7  |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |         |                       |
|-----|---|---------|-----------------------|
| 179 |   | 1575061 | Count coming in to IF |
| 179 | 1 | 10127   |                       |
| 218 | 1 | 1564934 |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |       |                       |
|-----|---|-------|-----------------------|
| 189 |   | 10127 | Count coming in to IF |
| 189 | 1 | 9679  |                       |
| 189 | 2 | 448   |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |       |                       |
|-----|---|-------|-----------------------|
| 192 |   | 10127 | Count coming in to IF |
| 192 | 1 | 5368  |                       |
|     |   | 4759  | All False Count       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |      |                       |
|-----|---|------|-----------------------|
| 193 |   | 5368 | Count coming in to IF |
| 193 | 1 | 224  |                       |
| 195 | 1 | 5144 |                       |

Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----

|     |   |      |                       |
|-----|---|------|-----------------------|
| 199 |   | 5368 | Count coming in to IF |
| 199 | 1 | 621  |                       |
|     |   | 4747 | All False Count       |

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  205                                10127    Count coming in to IF
  205              1                  4759
                                5368    All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  206                                4759    Count coming in to IF
  206              1                  4759
                                ***0***    All False Count
```

Branch totals: 1 hit of 2 branches = 50.00%

```
-----IF Branch-----
  213                                10127    Count coming in to IF
  213              1                   12
                                10115    All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  225                                104819    Count coming in to IF
  225              1                   621
  240              1                  104198
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  234                                621    Count coming in to IF
  234              1                   231
  237              1                   390
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  248                                7711226    Count coming in to IF
  248              1                  27173
  254              1                  7684053
```

Branch totals: 2 hits of 2 branches = 100.00%

```
-----IF Branch-----
  250                                27173    Count coming in to IF
  250              1                   231
                                26942    All False Count
```

Branch totals: 2 hits of 2 branches = 100.00%

Condition Coverage:

| Enabled Coverage | Bins | Covered | Misses | Coverage |
|------------------|------|---------|--------|----------|
| -----            | ---- | ----    | -----  | -----    |
| Conditions       | 19   | 18      | 1      | 94.73%   |

# =====Condition Details=====

Condition Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core --

File ./golden\_rtl/spi\_core.sv

-----Focused Condition View-----

Line 80 Item 1 (xfer\_width == 0)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 81 Item 1 (xfer\_width == 1)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 104 Item 1 (total\_bits == 32)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 149 Item 1 ((~tx\_empty && cfg\_mstr) && (ss\_n\_drive != 15))  
Condition totals: 3 of 3 input terms covered = 100.00%

-----Focused Condition View-----

Line 159 Item 1 (cfg\_width == 0)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 160 Item 1 (cfg\_width == 1)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 179 Item 1 (sclk\_cnt == (half\_period - 1))  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 199 Item 1 (bit\_cnt == 1)  
Condition totals: 1 of 1 input term covered = 100.00%

-----Focused Condition View-----

Line 206 Item 1 (bit\_cnt > 0)  
Condition totals: 0 of 1 input term covered = 0.00%

| Input Term    | Covered | Reason for no coverage | Hint     |
|---------------|---------|------------------------|----------|
| (bit_cnt > 0) | N       | '_0' not hit           | Hit '_0' |

| Rows:  | Hits    | FEC Target      | Non-masking condition(s) |
|--------|---------|-----------------|--------------------------|
| Row 1: | ***0*** | (bit_cnt > 0)_0 | -                        |
| Row 2: | 9       | (bit_cnt > 0)_1 | -                        |



```

-----Focused Condition View-----
Line      213 Item    1 ((cpha && leading) && (bit_cnt == width_bits))
Condition totals: 3 of 3 input terms covered = 100.00%

```

```

-----Focused Condition View-----
Line      225 Item    1 (sclk_cnt == (half_period - 1))
Condition totals: 1 of 1 input term covered = 100.00%

```

```

-----Focused Condition View-----
Line      234 Item    1 (~tx_empty && (cfg_delay != 0))
Condition totals: 2 of 2 input terms covered = 100.00%

```

```

-----Focused Condition View-----
Line      248 Item    1 (sclk_cnt == (half_period - 1))
Condition totals: 1 of 1 input term covered = 100.00%

```

```

-----Focused Condition View-----
Line      250 Item    1 (gap_cnt == 1)
Condition totals: 1 of 1 input term covered = 100.00%

```

#### Expression Coverage:

| Enabled Coverage | Bins | Covered | Misses | Coverage |
|------------------|------|---------|--------|----------|
| -----            | ---- | ----    | -----  | -----    |
| Expressions      | 9    | 9       | 0      | 100.00%  |

#### =====Expression Details=====

Expression Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core --

File ./golden\_rtl/spi\_core.sv

```

-----Focused Expression View-----
Line      87 Item    1 (state != S_IDLE)
Expression totals: 1 of 1 input term covered = 100.00%

```

```

-----Focused Expression View (Bimodal)-----
Line      89 Item    1 (cfg_loopback? MOSI: MISO)
Expression totals: 3 of 3 input terms covered = 100.00%

```

```

-----Focused Expression View (Bimodal)-----
Line     166 Item    1 (cfg_lsb_first? tx_word[0]:
tx_word[cfg_width==0?7:cfg_width==1?15:31])
Expression totals: 3 of 3 input terms covered = 100.00%

```

```

-----Focused Expression View (Bimodal)-----
Line     189 Item    1 (~cpha? leading: ~leading)
Expression totals: 2 of 2 input terms covered = 100.00%

```

#### FSM Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| -----            | ---- | ---- | -----  | -----    |
| FSM States       | 4    | 4    | 0      | 100.00%  |
| FSM Transitions  | 6    | 5    | 1      | 83.33%   |

=====FSM Details=====

FSM Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core --

FSM\_ID: state

Current State Object : state

-----  
State Value MapInfo :  
-----

| Line | State Name | Value |
|------|------------|-------|
| ---- | -----      | ----- |
| 145  | S_IDLE     | 0     |
| 178  | S_SHIFT    | 1     |
| 224  | S_FINISH   | 2     |
| 246  | S_GAP      | 3     |

Covered States :  
-----

| State    | Hit_count |
|----------|-----------|
| -----    | -----     |
| S_IDLE   | 1038      |
| S_SHIFT  | 621       |
| S_FINISH | 621       |
| S_GAP    | 231       |

Covered Transitions :  
-----

| Line | Trans_ID | Hit_count | Transition          |
|------|----------|-----------|---------------------|
| ---- | -----    | -----     | -----               |
| 172  | 0        | 621       | S_IDLE -> S_SHIFT   |
| 200  | 1        | 621       | S_SHIFT -> S_FINISH |
| 238  | 3        | 390       | S_FINISH -> S_IDLE  |
| 236  | 4        | 231       | S_FINISH -> S_GAP   |
| 251  | 5        | 231       | S_GAP -> S_IDLE     |

Uncovered Transitions :  
-----

| Line | Trans_ID | Transition        |
|------|----------|-------------------|
| ---- | -----    | -----             |
| 136  | 2        | S_SHIFT -> S_IDLE |

| Summary         | Bins | Hits | Misses | Coverage |
|-----------------|------|------|--------|----------|
| -----           | ---- | ---- | -----  | -----    |
| FSM States      | 4    | 4    | 0      | 100.00%  |
| FSM Transitions | 6    | 5    | 1      | 83.33%   |

Statement Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
|------------------|------|------|--------|----------|

|            |      |      |       |        |
|------------|------|------|-------|--------|
| -----      | ---- | ---- | ----- | -----  |
| Statements | 77   | 76   | 1     | 98.70% |

=====Statement Details=====

Statement Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core --

| Line                          | Item | Count   | Source |
|-------------------------------|------|---------|--------|
| ----                          | ---- | -----   | -----  |
| File ./golden_rtl/spi_core.sv |      |         |        |
| 80                            | 1    | 37      |        |
| 85                            | 1    | 167     |        |
| 87                            | 1    | 2114    |        |
| 89                            | 1    | 3956    |        |
| 97                            | 1    | 224     |        |
| 99                            | 1    | 4547    |        |
| 104                           | 1    | 621     |        |
| 111                           | 1    | 9393529 |        |
| 113                           | 1    | 30      |        |
| 114                           | 1    | 30      |        |
| 115                           | 1    | 30      |        |
| 116                           | 1    | 30      |        |
| 117                           | 1    | 30      |        |
| 118                           | 1    | 30      |        |
| 119                           | 1    | 30      |        |
| 120                           | 1    | 30      |        |
| 121                           | 1    | 30      |        |
| 122                           | 1    | 30      |        |
| 123                           | 1    | 30      |        |
| 124                           | 1    | 30      |        |
| 125                           | 1    | 30      |        |
| 126                           | 1    | 30      |        |
| 127                           | 1    | 30      |        |
| 128                           | 1    | 30      |        |
| 129                           | 1    | 30      |        |
| 131                           | 1    | 9393499 |        |
| 132                           | 1    | 9393499 |        |
| 133                           | 1    | 9393499 |        |
| 136                           | 1    | 387     |        |
| 137                           | 1    | 387     |        |
| 138                           | 1    | 387     |        |
| 139                           | 1    | 387     |        |
| 140                           | 1    | 387     |        |
| 141                           | 1    | 387     |        |
| 146                           | 1    | 2006    |        |
| 151                           | 1    | 621     |        |
| 152                           | 1    | 621     |        |
| 153                           | 1    | 621     |        |
| 154                           | 1    | 621     |        |
| 156                           | 1    | 621     |        |

|     |   |         |
|-----|---|---------|
| 157 | 1 | 621     |
| 159 | 1 | 621     |
| 161 | 1 | 621     |
| 162 | 1 | 621     |
| 166 | 1 | 609     |
| 171 | 1 | 621     |
| 172 | 1 | 621     |
| 173 | 1 | 621     |
| 180 | 1 | 10127   |
| 181 | 1 | 10127   |
| 182 | 1 | 10127   |
| 188 | 1 | 10127   |
| 189 | 1 | 10127   |
| 190 | 1 | 10127   |
| 194 | 1 | 224     |
| 196 | 1 | 5144    |
| 200 | 1 | 621     |
| 202 | 1 | 5368    |
| 207 | 1 | 4759    |
| 214 | 1 | 12      |
| 219 | 1 | 1564934 |
| 226 | 1 | 621     |
| 227 | 1 | 621     |
| 228 | 1 | 621     |
| 230 | 1 | 621     |
| 231 | 1 | 621     |
| 232 | 1 | 621     |
| 235 | 1 | 231     |
| 236 | 1 | 231     |
| 238 | 1 | 390     |
| 241 | 1 | 104198  |
| 247 | 1 | 7711226 |
| 249 | 1 | 27173   |
| 251 | 1 | 231     |
| 253 | 1 | 27173   |
| 255 | 1 | 7684053 |
| 259 | 1 | ***0*** |

Toggle Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| -----            | ---- | ---- | -----  | -----    |
| Toggles          | 512  | 443  | 69     | 86.52%   |

=====Toggle Details=====

Toggle Coverage for instance /tb\_top/u\_wrap/u\_dut/u\_core --

| "Coverage" | Node | 1H->0L | 0L->1H |
|------------|------|--------|--------|
|------------|------|--------|--------|

-----

|       |       |                    |           |    |
|-------|-------|--------------------|-----------|----|
|       |       | PRESETn            | 0         | 10 |
| 50.00 |       | cpol               | 0         | 1  |
| 50.00 |       | gap_cnt[8]         | 0         | 0  |
| 0.00  |       | half_period[11-16] | 0         | 0  |
| 0.00  |       | rx_push_data[20]   | 0         | 2  |
| 50.00 |       | rx_push_data[22]   | 0         | 0  |
| 0.00  |       | rx_push_data[24]   | 0         | 1  |
| 50.00 |       | sclk_cnt[11-16]    | 0         | 0  |
| 0.00  |       | sh_rx[22]          | 0         | 0  |
| 0.00  |       | sh_rx[24]          | 0         | 1  |
| 50.00 |       | sh_tx[17]          | 0         | 0  |
| 0.00  |       | sh_tx[20]          | 0         | 2  |
| 50.00 |       | sh_tx[22]          | 0         | 0  |
| 0.00  |       | sh_tx[24]          | 0         | 0  |
| 0.00  |       | sh_tx[29]          | 0         | 0  |
| 0.00  |       | state              | ENUM type |    |
| Value | Count | tx_word[17]        | 0         | 0  |
| 0.00  |       | tx_word[22]        | 0         | 0  |
| 0.00  |       | tx_word[24]        | 0         | 0  |
| 0.00  |       | tx_word[29]        | 0         | 0  |
| 0.00  |       | width_bits[0-2]    | 0         | 0  |
| 0.00  |       | xfer_div[11-15]    | 0         | 0  |
| 0.00  |       | xfer_mode[1]       | 0         | 1  |
| 50.00 |       |                    |           |    |

Total Node Count = 258  
 Toggled Node Count = 220  
 Untoggled Node Count = 38

Toggle Coverage = 86.52% (443 of 512 bins)

=====  
=== Instance: /tb\_top/u\_wrap/u\_dut

=== Design Unit: work.spi\_master  
=====

Toggle Coverage:

| Enabled Coverage | Bins | Hits | Misses | Coverage |
|------------------|------|------|--------|----------|
| -----            | ---- | ---- | -----  | -----    |
| Toggles          | 384  | 363  | 21     | 94.53%   |

=====Toggle Details=====

Toggle Coverage for instance /tb\_top/u\_wrap/u\_dut --

| "Coverage" | Node             | 1H->0L | 0L->1H |
|------------|------------------|--------|--------|
| -----      |                  |        |        |
|            | PRDATA[22]       | 0      | 0      |
| 0.00       |                  |        |        |
|            | PRDATA[24]       | 0      | 0      |
| 0.00       |                  |        |        |
|            | PREADY           | 0      | 0      |
| 0.00       |                  |        |        |
|            | PRESETn          | 0      | 10     |
| 50.00      |                  |        |        |
|            | PSLVERR          | 0      | 0      |
| 0.00       |                  |        |        |
|            | rx_push_data[20] | 0      | 2      |
| 50.00      |                  |        |        |
|            | rx_push_data[22] | 0      | 0      |
| 0.00       |                  |        |        |
|            | rx_push_data[24] | 0      | 1      |
| 50.00      |                  |        |        |
|            | tx_word[17]      | 0      | 0      |
| 0.00       |                  |        |        |
|            | tx_word[22]      | 0      | 0      |
| 0.00       |                  |        |        |
|            | tx_word[24]      | 0      | 0      |
| 0.00       |                  |        |        |
|            | tx_word[29]      | 0      | 0      |
| 0.00       |                  |        |        |

Total Node Count = 192

Toggled Node Count = 180

Untoggled Node Count = 12

Toggle Coverage = 94.53% (363 of 384 bins)

```

=== Instance: /tb_top_sv_unit
=== Design Unit: work.tb_top_sv_unit

```

|                     |     |     |    |         |
|---------------------|-----|-----|----|---------|
| Covergroups         | 15  | na  | na | 100.00% |
| Coverpoints/Crosses | 32  | na  | na | na      |
| Covergroup Bins     | 139 | 139 | 0  | 100.00% |

|                                                 |         |     |   |
|-------------------------------------------------|---------|-----|---|
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_config | 100.00% | 100 | - |
| Covered                                         |         |     |   |
| covered/total bins:                             | 42      | 42  | - |
| missing/total bins:                             | 0       | 42  | - |
| % Hit:                                          | 100.00% | 100 | - |
| Coverpoint cp_mode                              | 100.00% | 100 | - |
| Covered                                         |         |     |   |
| covered/total bins:                             | 4       | 4   | - |
| missing/total bins:                             | 0       | 4   | - |
| % Hit:                                          | 100.00% | 100 | - |
| bin modes[0]                                    | 178     | 1   | - |
| Covered                                         |         |     |   |
| bin modes[1]                                    | 8       | 1   | - |
| Covered                                         |         |     |   |
| bin modes[2]                                    | 8       | 1   | - |
| Covered                                         |         |     |   |
| bin modes[3]                                    | 9       | 1   | - |
| Covered                                         |         |     |   |
| Coverpoint cp_first                             | 100.00% | 100 | - |
| Covered                                         |         |     |   |
| covered/total bins:                             | 2       | 2   | - |
| missing/total bins:                             | 0       | 2   | - |
| % Hit:                                          | 100.00% | 100 | - |

|                                      |         |     |   |
|--------------------------------------|---------|-----|---|
| bin msb                              | 187     | 1   | - |
| Covered                              |         |     |   |
| bin lsb                              | 16      | 1   | - |
| Covered                              |         |     |   |
| Coverpoint cp_width                  | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 4       | 4   | - |
| missing/total bins:                  | 0       | 4   | - |
| % Hit:                               | 100.00% | 100 | - |
| bin w8                               | 173     | 1   | - |
| Covered                              |         |     |   |
| bin w16                              | 11      | 1   | - |
| Covered                              |         |     |   |
| bin w32                              | 11      | 1   | - |
| Covered                              |         |     |   |
| bin illegal_width                    | 8       | 1   | - |
| Covered                              |         |     |   |
| Cross cx_mode_width_order            | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 32      | 32  | - |
| missing/total bins:                  | 0       | 32  | - |
| % Hit:                               | 100.00% | 100 | - |
| Auto, Default and User Defined Bins: |         |     |   |
| bin <modes[3],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[2],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[1],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[0],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[3],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[2],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[1],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[0],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[3],w8,lsb>                | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[2],w8,lsb>                | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[1],w8,lsb>                | 1       | 1   | - |



|         |                                                                  |         |     |   |
|---------|------------------------------------------------------------------|---------|-----|---|
| Covered | bin <modes[0],w8,lsb>                                            | 1       | 1   | - |
| Covered | bin <modes[3],w32,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[2],w32,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[1],w32,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[3],w16,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[2],w16,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[1],w16,msb>                                           | 1       | 1   | - |
| Covered | bin <modes[3],w8,msb>                                            | 2       | 1   | - |
| Covered | bin <modes[2],w8,msb>                                            | 1       | 1   | - |
| Covered | bin <modes[1],w8,msb>                                            | 1       | 1   | - |
| Covered | bin <modes[0],w32,msb>                                           | 4       | 1   | - |
| Covered | bin <modes[0],w16,msb>                                           | 4       | 1   | - |
| Covered | bin <modes[3],illegal_width,lsb>                                 | 1       | 1   | - |
| Covered | bin <modes[2],illegal_width,lsb>                                 | 1       | 1   | - |
| Covered | bin <modes[1],illegal_width,lsb>                                 | 1       | 1   | - |
| Covered | bin <modes[0],illegal_width,lsb>                                 | 1       | 1   | - |
| Covered | bin <modes[3],illegal_width,msb>                                 | 1       | 1   | - |
| Covered | bin <modes[2],illegal_width,msb>                                 | 1       | 1   | - |
| Covered | bin <modes[1],illegal_width,msb>                                 | 1       | 1   | - |
| Covered | bin <modes[0],illegal_width,msb>                                 | 1       | 1   | - |
| Covered | bin <modes[0],w8,msb>                                            | 165     | 1   | - |
| Covered | Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_config | 100.00% | 100 | - |
| Covered | covered/total bins:                                              | 42      | 42  | - |
|         | missing/total bins:                                              | 0       | 42  | - |

|                           |         |     |   |
|---------------------------|---------|-----|---|
| % Hit:                    | 100.00% | 100 | - |
| Coverpoint cp_mode        | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 4       | 4   | - |
| missing/total bins:       | 0       | 4   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin modes[0]              | 178     | 1   | - |
| Covered                   |         |     |   |
| bin modes[1]              | 8       | 1   | - |
| Covered                   |         |     |   |
| bin modes[2]              | 8       | 1   | - |
| Covered                   |         |     |   |
| bin modes[3]              | 9       | 1   | - |
| Covered                   |         |     |   |
| Coverpoint cp_first       | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 2       | 2   | - |
| missing/total bins:       | 0       | 2   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin msb                   | 187     | 1   | - |
| Covered                   |         |     |   |
| bin lsb                   | 16      | 1   | - |
| Covered                   |         |     |   |
| Coverpoint cp_width       | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 4       | 4   | - |
| missing/total bins:       | 0       | 4   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin w8                    | 173     | 1   | - |
| Covered                   |         |     |   |
| bin w16                   | 11      | 1   | - |
| Covered                   |         |     |   |
| bin w32                   | 11      | 1   | - |
| Covered                   |         |     |   |
| bin illegal_width         | 8       | 1   | - |
| Covered                   |         |     |   |
| Cross cx_mode_width_order | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 32      | 32  | - |

|                                      |                        |         |     |   |
|--------------------------------------|------------------------|---------|-----|---|
| missing/total bins:                  |                        | 0       | 32  | - |
| % Hit:                               |                        | 100.00% | 100 | - |
| Auto, Default and User Defined Bins: |                        |         |     |   |
| Covered                              | bin <modes[3],w32,lsb> | 1       | 1   | - |
|                                      | bin <modes[2],w32,lsb> | 1       | 1   | - |
|                                      | bin <modes[1],w32,lsb> | 1       | 1   | - |
|                                      | bin <modes[0],w32,lsb> | 1       | 1   | - |
| Covered                              | bin <modes[3],w16,lsb> | 1       | 1   | - |
|                                      | bin <modes[2],w16,lsb> | 1       | 1   | - |
|                                      | bin <modes[1],w16,lsb> | 1       | 1   | - |
|                                      | bin <modes[0],w16,lsb> | 1       | 1   | - |
| Covered                              | bin <modes[3],w8,lsb>  | 1       | 1   | - |
|                                      | bin <modes[2],w8,lsb>  | 1       | 1   | - |
|                                      | bin <modes[1],w8,lsb>  | 1       | 1   | - |
|                                      | bin <modes[0],w8,lsb>  | 1       | 1   | - |
| Covered                              | bin <modes[3],w32,msb> | 1       | 1   | - |
|                                      | bin <modes[2],w32,msb> | 1       | 1   | - |
|                                      | bin <modes[1],w32,msb> | 1       | 1   | - |
|                                      | bin <modes[3],w16,msb> | 1       | 1   | - |
| Covered                              | bin <modes[2],w16,msb> | 1       | 1   | - |
|                                      | bin <modes[1],w16,msb> | 1       | 1   | - |
|                                      | bin <modes[3],w8,msb>  | 2       | 1   | - |
|                                      | bin <modes[2],w8,msb>  | 1       | 1   | - |
| Covered                              | bin <modes[1],w8,msb>  | 1       | 1   | - |
|                                      | bin <modes[0],w32,msb> | 4       | 1   | - |
| Covered                              | bin <modes[0],w16,msb> | 4       | 1   | - |

|                                                  |                                  |         |     |   |
|--------------------------------------------------|----------------------------------|---------|-----|---|
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[3],illegal_width,lsb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[2],illegal_width,lsb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[1],illegal_width,lsb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[0],illegal_width,lsb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[3],illegal_width,msb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[2],illegal_width,msb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[1],illegal_width,msb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[0],illegal_width,msb> | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin <modes[0],w8,msb>            | 165     | 1   | - |
| Covered                                          |                                  |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_clk_div |                                  | 100.00% | 100 | - |
| Covered                                          |                                  |         |     |   |
|                                                  | covered/total bins:              | 10      | 10  | - |
|                                                  |                                  |         |     |   |
|                                                  | missing/total bins:              | 0       | 10  | - |
|                                                  |                                  |         |     |   |
|                                                  | % Hit:                           | 100.00% | 100 | - |
|                                                  |                                  |         |     |   |
|                                                  | Coverpoint cp_clk_div            | 100.00% | 100 | - |
| Covered                                          |                                  |         |     |   |
|                                                  | covered/total bins:              | 10      | 10  | - |
|                                                  |                                  |         |     |   |
|                                                  | missing/total bins:              | 0       | 10  | - |
|                                                  |                                  |         |     |   |
|                                                  | % Hit:                           | 100.00% | 100 | - |
|                                                  |                                  |         |     |   |
|                                                  | bin div_zero                     | 2       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_one                      | 38      | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_small                    | 11      | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_typical                  | 44      | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_medium                   | 57      | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_255                      | 1       | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_large                    | 37      | 1   | - |
| Covered                                          |                                  |         |     |   |
|                                                  | bin div_1024                     | 7       | 1   | - |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| Covered                                                           |         |     |   |
| bin div_very_big                                                  | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_max                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_clk_div |         |     |   |
|                                                                   | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 10      | 10  | - |
| missing/total bins:                                               | 0       | 10  | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_clk_div                                             | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 10      | 10  | - |
| missing/total bins:                                               | 0       | 10  | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin div_zero                                                      | 2       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_one                                                       | 38      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_small                                                     | 11      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_typical                                                   | 44      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_medium                                                    | 57      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_255                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_large                                                     | 37      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_1024                                                      | 7       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_very_big                                                  | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_max                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_interrupt_sources        |         |     |   |
|                                                                   | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 5       | 5   | - |
| missing/total bins:                                               | 0       | 5   | - |
| % Hit:                                                            | 100.00% | 100 | - |

|                            |         |     |   |
|----------------------------|---------|-----|---|
| Coverpoint cp_tx_empty_irq | 100.00% | 100 | - |
| Covered                    |         |     |   |
| covered/total bins:        | 1       | 1   | - |
| missing/total bins:        | 0       | 1   | - |
| % Hit:                     | 100.00% | 100 | - |
| bin fired                  | 36      | 1   | - |
| Covered                    |         |     |   |
| Coverpoint cp_rx_full_irq  | 100.00% | 100 | - |
| Covered                    |         |     |   |
| covered/total bins:        | 1       | 1   | - |
| missing/total bins:        | 0       | 1   | - |
| % Hit:                     | 100.00% | 100 | - |
| bin fired                  | 4       | 1   | - |
| Covered                    |         |     |   |
| Coverpoint cp_tx_ovf_irq   | 100.00% | 100 | - |
| Covered                    |         |     |   |
| covered/total bins:        | 1       | 1   | - |
| missing/total bins:        | 0       | 1   | - |
| % Hit:                     | 100.00% | 100 | - |
| bin fired                  | 3       | 1   | - |
| Covered                    |         |     |   |
| Coverpoint cp_rx_ovf_irq   | 100.00% | 100 | - |
| Covered                    |         |     |   |
| covered/total bins:        | 1       | 1   | - |
| missing/total bins:        | 0       | 1   | - |
| % Hit:                     | 100.00% | 100 | - |
| bin fired                  | 5       | 1   | - |
| Covered                    |         |     |   |
| Coverpoint cp_done_irq     | 100.00% | 100 | - |
| Covered                    |         |     |   |
| covered/total bins:        | 1       | 1   | - |
| missing/total bins:        | 0       | 1   | - |
| % Hit:                     | 100.00% | 100 | - |
| bin fired                  | 36      | 1   | - |

|                                                                             |         |     |   |
|-----------------------------------------------------------------------------|---------|-----|---|
| Covered                                                                     |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_interrupt_sources | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 5       | 5   | - |
| missing/total bins:                                                         | 0       | 5   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_irq                                                  | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 36      | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_rx_full_irq                                                   | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 4       | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_tx_ovf_irq                                                    | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 3       | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_rx_ovf_irq                                                    | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 5       | 1   | - |
| Covered                                                                     |         |     |   |

|                                                  |         |     |   |
|--------------------------------------------------|---------|-----|---|
| Coverpoint cp_done_irq                           | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| bin fired                                        | 36      | 1   | - |
| Covered                                          |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_irq_w1c | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 5       | 5   | - |
| missing/total bins:                              | 0       | 5   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_w1c                       | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_rx_full_w1c                        | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_tx_ovf_w1c                         | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_rx_ovf_w1c                         | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |



|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| Coverpoint cp_done_w1c                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_irq_w1c | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 5       | 5   | - |
| missing/total bins:                                               | 0       | 5   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_w1c                                        | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| Coverpoint cp_rx_full_w1c                                         | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| Coverpoint cp_tx_ovf_w1c                                          | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |
| Coverpoint cp_rx_ovf_w1c                                          | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |

|                                                     |         |     |   |
|-----------------------------------------------------|---------|-----|---|
| missing/total bins:                                 | 0       | 1   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin cleared                                         | 1       | 1   | - |
| Covered                                             |         |     |   |
| Coverpoint cp_done_w1c                              | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 1       | 1   | - |
| missing/total bins:                                 | 0       | 1   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin cleared                                         | 1       | 1   | - |
| Covered                                             |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_irq_masked | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 10      | 10  | - |
| missing/total bins:                                 | 0       | 10  | - |
| % Hit:                                              | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_masked                       | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 2       | 2   | - |
| missing/total bins:                                 | 0       | 2   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin fired_unmasked                                  | 26      | 1   | - |
| Covered                                             |         |     |   |
| bin fired_masked                                    | 10      | 1   | - |
| Covered                                             |         |     |   |
| Coverpoint cp_rx_full_masked                        | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 2       | 2   | - |
| missing/total bins:                                 | 0       | 2   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin fired_unmasked                                  | 2       | 1   | - |
| Covered                                             |         |     |   |
| bin fired_masked                                    | 2       | 1   | - |
| Covered                                             |         |     |   |

|                                                                      |         |     |   |
|----------------------------------------------------------------------|---------|-----|---|
| Coverpoint cp_tx_ovf_masked                                          | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 2       | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 1       | 1   | - |
| Covered                                                              |         |     |   |
| Coverpoint cp_rx_ovf_masked                                          | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 2       | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 3       | 1   | - |
| Covered                                                              |         |     |   |
| Coverpoint cp_done_masked                                            | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 28      | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 8       | 1   | - |
| Covered                                                              |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_irq_masked | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 10      | 10  | - |
| missing/total bins:                                                  | 0       | 10  | - |
| % Hit:                                                               | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_masked                                        | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |

|                              |         |     |   |
|------------------------------|---------|-----|---|
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 26      | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 10      | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_rx_full_masked | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 2       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_tx_ovf_masked  | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 1       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_rx_ovf_masked  | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 3       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_done_masked    | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |

|                                                                 |         |     |   |
|-----------------------------------------------------------------|---------|-----|---|
| bin fired_unmasked                                              | 28      | 1   | - |
| Covered                                                         |         |     |   |
| bin fired_masked                                                | 8       | 1   | - |
| Covered                                                         |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_delay                  | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| Coverpoint cp_delay                                             | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| bin delay_zero                                                  | 47      | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_min                                                   | 4       | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_small                                                 | 6       | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_medium                                                | 52      | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_large                                                 | 58      | 1   | - |
| Covered                                                         |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_delay | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| Coverpoint cp_delay                                             | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| bin delay_zero                                                  | 47      | 1   | - |
| Covered                                                         |         |     |   |

|                                                                     |         |     |   |
|---------------------------------------------------------------------|---------|-----|---|
| bin delay_min                                                       | 4       | 1   | - |
| Covered                                                             |         |     |   |
| bin delay_small                                                     | 6       | 1   | - |
| Covered                                                             |         |     |   |
| bin delay_medium                                                    | 52      | 1   | - |
| Covered                                                             |         |     |   |
| bin delay_large                                                     | 58      | 1   | - |
| Covered                                                             |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_ss_select                  | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 11      | 11  | - |
| missing/total bins:                                                 | 0       | 11  | - |
| % Hit:                                                              | 100.00% | 100 | - |
| Coverpoint cp_ss_en                                                 | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 11      | 11  | - |
| missing/total bins:                                                 | 0       | 11  | - |
| % Hit:                                                              | 100.00% | 100 | - |
| bin slave0_only                                                     | 44      | 1   | - |
| Covered                                                             |         |     |   |
| bin slave1_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin slave2_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin slave3_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin no_slave                                                        | 35      | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[3]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[5]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[6]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[9]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[10]                                                 | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[12]                                                 | 2       | 1   | - |
| Covered                                                             |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_ss_select | 100.00% | 100 | - |
| Covered                                                             |         |     |   |

|                                                   |         |     |   |
|---------------------------------------------------|---------|-----|---|
| covered/total bins:                               | 11      | 11  | - |
| missing/total bins:                               | 0       | 11  | - |
| % Hit:                                            | 100.00% | 100 | - |
| Coverpoint cp_ss_en                               | 100.00% | 100 | - |
| Covered                                           |         |     |   |
| covered/total bins:                               | 11      | 11  | - |
| missing/total bins:                               | 0       | 11  | - |
| % Hit:                                            | 100.00% | 100 | - |
| bin slave0_only                                   | 44      | 1   | - |
| Covered                                           |         |     |   |
| bin slave1_only                                   | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin slave2_only                                   | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin slave3_only                                   | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin no_slave                                      | 35      | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[3]                                | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[5]                                | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[6]                                | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[9]                                | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[10]                               | 1       | 1   | - |
| Covered                                           |         |     |   |
| bin multi_slave[12]                               | 2       | 1   | - |
| Covered                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_loopback | 100.00% | 100 | - |
| Covered                                           |         |     |   |
| covered/total bins:                               | 11      | 11  | - |
| missing/total bins:                               | 0       | 11  | - |
| % Hit:                                            | 100.00% | 100 | - |
| Coverpoint cp_loopback                            | 100.00% | 100 | - |
| Covered                                           |         |     |   |
| covered/total bins:                               | 2       | 2   | - |
| missing/total bins:                               | 0       | 2   | - |

|                                                                    |         |     |   |
|--------------------------------------------------------------------|---------|-----|---|
| % Hit:                                                             | 100.00% | 100 | - |
| bin off                                                            | 7       | 1   | - |
| Covered                                                            |         |     |   |
| bin on                                                             | 38      | 1   | - |
| Covered                                                            |         |     |   |
| Coverpoint cp_width                                                | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 3       | 3   | - |
| missing/total bins:                                                | 0       | 3   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| bin w8                                                             | 23      | 1   | - |
| Covered                                                            |         |     |   |
| bin w16                                                            | 11      | 1   | - |
| Covered                                                            |         |     |   |
| bin w32                                                            | 11      | 1   | - |
| Covered                                                            |         |     |   |
| Cross cx_loopback_width                                            | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 6       | 6   | - |
| missing/total bins:                                                | 0       | 6   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| Auto, Default and User Defined Bins:                               |         |     |   |
| bin <on,w32>                                                       | 9       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w32>                                                      | 2       | 1   | - |
| Covered                                                            |         |     |   |
| bin <on,w16>                                                       | 9       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w16>                                                      | 2       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w8>                                                       | 3       | 1   | - |
| Covered                                                            |         |     |   |
| bin <on,w8>                                                        | 20      | 1   | - |
| Covered                                                            |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_loopback | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 11      | 11  | - |
| missing/total bins:                                                | 0       | 11  | - |
| % Hit:                                                             | 100.00% | 100 | - |



|                                                  |         |     |   |
|--------------------------------------------------|---------|-----|---|
| Coverpoint cp_loopback                           | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 2       | 2   | - |
| missing/total bins:                              | 0       | 2   | - |
| % Hit:                                           | 100.00% | 100 | - |
| bin off                                          | 7       | 1   | - |
| Covered                                          |         |     |   |
| bin on                                           | 38      | 1   | - |
| Covered                                          |         |     |   |
| Coverpoint cp_width                              | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 3       | 3   | - |
| missing/total bins:                              | 0       | 3   | - |
| % Hit:                                           | 100.00% | 100 | - |
| bin w8                                           | 23      | 1   | - |
| Covered                                          |         |     |   |
| bin w16                                          | 11      | 1   | - |
| Covered                                          |         |     |   |
| bin w32                                          | 11      | 1   | - |
| Covered                                          |         |     |   |
| Cross cx_loopback_width                          | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 6       | 6   | - |
| missing/total bins:                              | 0       | 6   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Auto, Default and User Defined Bins:             |         |     |   |
| bin <on,w32>                                     | 9       | 1   | - |
| Covered                                          |         |     |   |
| bin <off,w32>                                    | 2       | 1   | - |
| Covered                                          |         |     |   |
| bin <on,w16>                                     | 9       | 1   | - |
| Covered                                          |         |     |   |
| bin <off,w16>                                    | 2       | 1   | - |
| Covered                                          |         |     |   |
| bin <off,w8>                                     | 3       | 1   | - |
| Covered                                          |         |     |   |
| bin <on,w8>                                      | 20      | 1   | - |
| Covered                                          |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_tx_fifo | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 8       | 8   | - |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_tx_fifo | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin tx_empty                                                      | 40      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_one_word                                                   | 40      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_partial                                                    | 88      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_mid                                                        | 22      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_almost_full                                                | 11      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_full                                                       | 7       | 1   | - |
| Covered                                                           |         |     |   |
| bin empty_to_full                                                 | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin full_to_empty                                                 | 5       | 1   | - |
| Covered                                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_rx_fifo                  | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_rx_count<br>Covered                                 | 100.00% | 100 | - |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_rx_fifo | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_rx_count<br>Covered                                 | 100.00% | 100 | - |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin rx_empty<br>Covered                                           | 40      | 1   | - |
| bin rx_one_word<br>Covered                                        | 13      | 1   | - |
| bin rx_partial<br>Covered                                         | 17      | 1   | - |
| bin rx_almost_full<br>Covered                                     | 6       | 1   | - |
| bin rx_full<br>Covered                                            | 6       | 1   | - |
| bin empty_to_full<br>Covered                                      | 5       | 1   | - |
| bin full_to_empty<br>Covered                                      | 5       | 1   | - |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_busy<br>Covered          | 100.00% | 100 | - |
| covered/total bins:                                               | 2       | 2   | - |
| missing/total bins:                                               | 0       | 2   | - |
| % Hit:                                                            | 100.00% | 100 | - |

|                                                                |         |     |   |
|----------------------------------------------------------------|---------|-----|---|
| Coverpoint cp_busy                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| bin idle                                                       | 200     | 1   | - |
| Covered                                                        |         |     |   |
| bin active                                                     | 137     | 1   | - |
| Covered                                                        |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_busy | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| Coverpoint cp_busy                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| bin idle                                                       | 200     | 1   | - |
| Covered                                                        |         |     |   |
| bin active                                                     | 137     | 1   | - |
| Covered                                                        |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_transfer_done         | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 1       | 1   | - |
| missing/total bins:                                            | 0       | 1   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| Coverpoint cp_done                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 1       | 1   | - |
| missing/total bins:                                            | 0       | 1   | - |
| % Hit:                                                         | 100.00% | 100 | - |

|                                                                         |         |     |   |
|-------------------------------------------------------------------------|---------|-----|---|
| bin done                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_transfer_done | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 1       | 1   | - |
| missing/total bins:                                                     | 0       | 1   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| Coverpoint cp_done                                                      | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 1       | 1   | - |
| missing/total bins:                                                     | 0       | 1   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| bin done                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reg_written                    | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 7       | 7   | - |
| missing/total bins:                                                     | 0       | 7   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| Coverpoint cp_reg_written                                               | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 7       | 7   | - |
| missing/total bins:                                                     | 0       | 7   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| bin ctrl                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin tx_data                                                             | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin clk_div                                                             | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin ss_ctrl                                                             | 85      | 1   | - |
| Covered                                                                 |         |     |   |
| bin int_en                                                              | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin int_stat                                                            | 1       | 1   | - |

|                                                                       |         |     |   |
|-----------------------------------------------------------------------|---------|-----|---|
| Covered                                                               |         |     |   |
| bin delay_reg                                                         | 43      | 1   | - |
| Covered                                                               |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reg_written | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 7       | 7   | - |
| missing/total bins:                                                   | 0       | 7   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| Coverpoint cp_reg_written                                             | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 7       | 7   | - |
| missing/total bins:                                                   | 0       | 7   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| bin ctrl                                                              | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin tx_data                                                           | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin clk_div                                                           | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin ss_ctrl                                                           | 85      | 1   | - |
| Covered                                                               |         |     |   |
| bin int_en                                                            | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin int_stat                                                          | 1       | 1   | - |
| Covered                                                               |         |     |   |
| bin delay_reg                                                         | 43      | 1   | - |
| Covered                                                               |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reg_read                     | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 8       | 8   | - |
| missing/total bins:                                                   | 0       | 8   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| Coverpoint cp_reg_read                                                | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 8       | 8   | - |
| missing/total bins:                                                   | 0       | 8   | - |
| % Hit:                                                                | 100.00% | 100 | - |

|                                                                    |         |     |   |
|--------------------------------------------------------------------|---------|-----|---|
| bin ctrl                                                           | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin status                                                         | 40      | 1   | - |
| Covered                                                            |         |     |   |
| bin rx_data                                                        | 48      | 1   | - |
| Covered                                                            |         |     |   |
| bin clk_div                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin ss_ctrl                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_en                                                         | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_stat                                                       | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin delay_reg                                                      | 6       | 1   | - |
| Covered                                                            |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reg_read |         |     |   |
|                                                                    | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 8       | 8   | - |
| missing/total bins:                                                | 0       | 8   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| Coverpoint cp_reg_read                                             | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 8       | 8   | - |
| missing/total bins:                                                | 0       | 8   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| bin ctrl                                                           | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin status                                                         | 40      | 1   | - |
| Covered                                                            |         |     |   |
| bin rx_data                                                        | 48      | 1   | - |
| Covered                                                            |         |     |   |
| bin clk_div                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin ss_ctrl                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_en                                                         | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_stat                                                       | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin delay_reg                                                      | 6       | 1   | - |
| Covered                                                            |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reset_values              |         |     |   |

|                                                                        |         |     |   |
|------------------------------------------------------------------------|---------|-----|---|
|                                                                        | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| Coverpoint cp_reset_read                                               | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| bin ctrl_reset                                                         | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin status_reset                                                       | 40      | 1   | - |
| Covered                                                                |         |     |   |
| bin clkdiv_reset                                                       | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin ssctrl_reset                                                       | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin inten_reset                                                        | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin intstat_reset                                                      | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin delay_reset                                                        | 6       | 1   | - |
| Covered                                                                |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reset_values | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| Coverpoint cp_reset_read                                               | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| bin ctrl_reset                                                         | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin status_reset                                                       | 40      | 1   | - |



|                   |   |   |   |
|-------------------|---|---|---|
| Covered           |   |   |   |
| bin clkdiv_reset  | 6 | 1 | - |
| Covered           |   |   |   |
| bin ssctrl_reset  | 6 | 1 | - |
| Covered           |   |   |   |
| bin inten_reset   | 6 | 1 | - |
| Covered           |   |   |   |
| bin intstat_reset | 6 | 1 | - |
| Covered           |   |   |   |
| bin delay_reset   | 6 | 1 | - |
| Covered           |   |   |   |

COVERGROUP COVERAGE:

| Covergroup                                      | Metric  | Goal | Bins |
|-------------------------------------------------|---------|------|------|
| Status                                          |         |      |      |
| -----                                           |         |      |      |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_config | 100.00% | 100  | -    |
| Covered                                         |         |      |      |
| covered/total bins:                             | 42      | 42   | -    |
| missing/total bins:                             | 0       | 42   | -    |
| % Hit:                                          | 100.00% | 100  | -    |
| Coverpoint cp_mode                              | 100.00% | 100  | -    |
| Covered                                         |         |      |      |
| covered/total bins:                             | 4       | 4    | -    |
| missing/total bins:                             | 0       | 4    | -    |
| % Hit:                                          | 100.00% | 100  | -    |
| bin modes[0]                                    | 178     | 1    | -    |
| Covered                                         |         |      |      |
| bin modes[1]                                    | 8       | 1    | -    |
| Covered                                         |         |      |      |
| bin modes[2]                                    | 8       | 1    | -    |
| Covered                                         |         |      |      |
| bin modes[3]                                    | 9       | 1    | -    |
| Covered                                         |         |      |      |
| Coverpoint cp_first                             | 100.00% | 100  | -    |
| Covered                                         |         |      |      |
| covered/total bins:                             | 2       | 2    | -    |
| missing/total bins:                             | 0       | 2    | -    |

|                                      |         |     |   |
|--------------------------------------|---------|-----|---|
| % Hit:                               | 100.00% | 100 | - |
| bin msb                              | 187     | 1   | - |
| Covered                              |         |     |   |
| bin lsb                              | 16      | 1   | - |
| Covered                              |         |     |   |
| Coverpoint cp_width                  | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 4       | 4   | - |
| missing/total bins:                  | 0       | 4   | - |
| % Hit:                               | 100.00% | 100 | - |
| bin w8                               | 173     | 1   | - |
| Covered                              |         |     |   |
| bin w16                              | 11      | 1   | - |
| Covered                              |         |     |   |
| bin w32                              | 11      | 1   | - |
| Covered                              |         |     |   |
| bin illegal_width                    | 8       | 1   | - |
| Covered                              |         |     |   |
| Cross cx_mode_width_order            | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 32      | 32  | - |
| missing/total bins:                  | 0       | 32  | - |
| % Hit:                               | 100.00% | 100 | - |
| Auto, Default and User Defined Bins: |         |     |   |
| bin <modes[3],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[2],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[1],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[0],w32,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[3],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[2],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[1],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[0],w16,lsb>               | 1       | 1   | - |
| Covered                              |         |     |   |
| bin <modes[3],w8,lsb>                | 1       | 1   | - |
| Covered                              |         |     |   |

|                     |                                              |         |     |   |
|---------------------|----------------------------------------------|---------|-----|---|
| Covered             | bin <modes[2],w8,lsb>                        | 1       | 1   | - |
| Covered             | bin <modes[1],w8,lsb>                        | 1       | 1   | - |
| Covered             | bin <modes[0],w8,lsb>                        | 1       | 1   | - |
| Covered             | bin <modes[3],w32,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[2],w32,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[1],w32,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[3],w16,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[2],w16,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[1],w16,msb>                       | 1       | 1   | - |
| Covered             | bin <modes[3],w8,msb>                        | 2       | 1   | - |
| Covered             | bin <modes[2],w8,msb>                        | 1       | 1   | - |
| Covered             | bin <modes[1],w8,msb>                        | 1       | 1   | - |
| Covered             | bin <modes[0],w32,msb>                       | 4       | 1   | - |
| Covered             | bin <modes[0],w16,msb>                       | 4       | 1   | - |
| Covered             | bin <modes[3],illegal_width,lsb>             | 1       | 1   | - |
| Covered             | bin <modes[2],illegal_width,lsb>             | 1       | 1   | - |
| Covered             | bin <modes[1],illegal_width,lsb>             | 1       | 1   | - |
| Covered             | bin <modes[0],illegal_width,lsb>             | 1       | 1   | - |
| Covered             | bin <modes[3],illegal_width,msb>             | 1       | 1   | - |
| Covered             | bin <modes[2],illegal_width,msb>             | 1       | 1   | - |
| Covered             | bin <modes[1],illegal_width,msb>             | 1       | 1   | - |
| Covered             | bin <modes[0],illegal_width,msb>             | 1       | 1   | - |
| Covered             | bin <modes[0],w8,msb>                        | 165     | 1   | - |
| Covergroup instance | \tb_top_sv_unit::spi_coverage_col::cg_config |         |     |   |
|                     |                                              | 100.00% | 100 | - |
| Covered             | covered/total bins:                          | 42      | 42  | - |

|                           |         |     |   |
|---------------------------|---------|-----|---|
| missing/total bins:       | 0       | 42  | - |
| % Hit:                    | 100.00% | 100 | - |
| Coverpoint cp_mode        | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 4       | 4   | - |
| missing/total bins:       | 0       | 4   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin modes[0]              | 178     | 1   | - |
| Covered                   |         |     |   |
| bin modes[1]              | 8       | 1   | - |
| Covered                   |         |     |   |
| bin modes[2]              | 8       | 1   | - |
| Covered                   |         |     |   |
| bin modes[3]              | 9       | 1   | - |
| Covered                   |         |     |   |
| Coverpoint cp_first       | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 2       | 2   | - |
| missing/total bins:       | 0       | 2   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin msb                   | 187     | 1   | - |
| Covered                   |         |     |   |
| bin lsb                   | 16      | 1   | - |
| Covered                   |         |     |   |
| Coverpoint cp_width       | 100.00% | 100 | - |
| Covered                   |         |     |   |
| covered/total bins:       | 4       | 4   | - |
| missing/total bins:       | 0       | 4   | - |
| % Hit:                    | 100.00% | 100 | - |
| bin w8                    | 173     | 1   | - |
| Covered                   |         |     |   |
| bin w16                   | 11      | 1   | - |
| Covered                   |         |     |   |
| bin w32                   | 11      | 1   | - |
| Covered                   |         |     |   |
| bin illegal_width         | 8       | 1   | - |
| Covered                   |         |     |   |
| Cross cx_mode_width_order | 100.00% | 100 | - |

|         |                                      |         |     |   |
|---------|--------------------------------------|---------|-----|---|
| Covered |                                      |         |     |   |
|         | covered/total bins:                  | 32      | 32  | - |
|         | missing/total bins:                  | 0       | 32  | - |
|         | % Hit:                               | 100.00% | 100 | - |
|         | Auto, Default and User Defined Bins: |         |     |   |
|         | bin <modes[3],w32,lsb>               | 1       | 1   | - |
| Covered | bin <modes[2],w32,lsb>               | 1       | 1   | - |
| Covered | bin <modes[1],w32,lsb>               | 1       | 1   | - |
| Covered | bin <modes[0],w32,lsb>               | 1       | 1   | - |
| Covered | bin <modes[3],w16,lsb>               | 1       | 1   | - |
| Covered | bin <modes[2],w16,lsb>               | 1       | 1   | - |
| Covered | bin <modes[1],w16,lsb>               | 1       | 1   | - |
| Covered | bin <modes[0],w16,lsb>               | 1       | 1   | - |
| Covered | bin <modes[3],w8,lsb>                | 1       | 1   | - |
| Covered | bin <modes[2],w8,lsb>                | 1       | 1   | - |
| Covered | bin <modes[1],w8,lsb>                | 1       | 1   | - |
| Covered | bin <modes[0],w8,lsb>                | 1       | 1   | - |
| Covered | bin <modes[3],w32,msb>               | 1       | 1   | - |
| Covered | bin <modes[2],w32,msb>               | 1       | 1   | - |
| Covered | bin <modes[1],w32,msb>               | 1       | 1   | - |
| Covered | bin <modes[3],w16,msb>               | 1       | 1   | - |
| Covered | bin <modes[2],w16,msb>               | 1       | 1   | - |
| Covered | bin <modes[1],w16,msb>               | 1       | 1   | - |
| Covered | bin <modes[3],w8,msb>                | 2       | 1   | - |
| Covered | bin <modes[2],w8,msb>                | 1       | 1   | - |
| Covered | bin <modes[1],w8,msb>                | 1       | 1   | - |
| Covered |                                      |         |     |   |

|         |                                                  |         |     |   |
|---------|--------------------------------------------------|---------|-----|---|
| Covered | bin <modes[0],w32,msb>                           | 4       | 1   | - |
| Covered | bin <modes[0],w16,msb>                           | 4       | 1   | - |
| Covered | bin <modes[3],illegal_width,lsb>                 | 1       | 1   | - |
| Covered | bin <modes[2],illegal_width,lsb>                 | 1       | 1   | - |
| Covered | bin <modes[1],illegal_width,lsb>                 | 1       | 1   | - |
| Covered | bin <modes[0],illegal_width,lsb>                 | 1       | 1   | - |
| Covered | bin <modes[3],illegal_width,msb>                 | 1       | 1   | - |
| Covered | bin <modes[2],illegal_width,msb>                 | 1       | 1   | - |
| Covered | bin <modes[1],illegal_width,msb>                 | 1       | 1   | - |
| Covered | bin <modes[0],illegal_width,msb>                 | 1       | 1   | - |
| Covered | bin <modes[0],w8,msb>                            | 165     | 1   | - |
| Covered | TYPE /tb_top_sv_unit/spi_coverage_col/cg_clk_div | 100.00% | 100 | - |
| Covered | covered/total bins:                              | 10      | 10  | - |
|         | missing/total bins:                              | 0       | 10  | - |
|         | % Hit:                                           | 100.00% | 100 | - |
| Covered | Coverpoint cp_clk_div                            | 100.00% | 100 | - |
| Covered | covered/total bins:                              | 10      | 10  | - |
|         | missing/total bins:                              | 0       | 10  | - |
|         | % Hit:                                           | 100.00% | 100 | - |
| Covered | bin div_zero                                     | 2       | 1   | - |
| Covered | bin div_one                                      | 38      | 1   | - |
| Covered | bin div_small                                    | 11      | 1   | - |
| Covered | bin div_typical                                  | 44      | 1   | - |
| Covered | bin div_medium                                   | 57      | 1   | - |
| Covered | bin div_255                                      | 1       | 1   | - |
| Covered |                                                  |         |     |   |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| bin div_large                                                     | 37      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_1024                                                      | 7       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_very_big                                                  | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_max                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_clk_div |         |     |   |
|                                                                   | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 10      | 10  | - |
| missing/total bins:                                               | 0       | 10  | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_clk_div                                             | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 10      | 10  | - |
| missing/total bins:                                               | 0       | 10  | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin div_zero                                                      | 2       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_one                                                       | 38      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_small                                                     | 11      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_typical                                                   | 44      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_medium                                                    | 57      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_255                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_large                                                     | 37      | 1   | - |
| Covered                                                           |         |     |   |
| bin div_1024                                                      | 7       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_very_big                                                  | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin div_max                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_interrupt_sources        |         |     |   |
|                                                                   | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 5       | 5   | - |

|                                       |         |     |   |
|---------------------------------------|---------|-----|---|
| missing/total bins:                   | 0       | 5   | - |
| % Hit:                                | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_irq<br>Covered | 100.00% | 100 | - |
| covered/total bins:                   | 1       | 1   | - |
| missing/total bins:                   | 0       | 1   | - |
| % Hit:                                | 100.00% | 100 | - |
| bin fired<br>Covered                  | 36      | 1   | - |
| Coverpoint cp_rx_full_irq<br>Covered  | 100.00% | 100 | - |
| covered/total bins:                   | 1       | 1   | - |
| missing/total bins:                   | 0       | 1   | - |
| % Hit:                                | 100.00% | 100 | - |
| bin fired<br>Covered                  | 4       | 1   | - |
| Coverpoint cp_tx_ovf_irq<br>Covered   | 100.00% | 100 | - |
| covered/total bins:                   | 1       | 1   | - |
| missing/total bins:                   | 0       | 1   | - |
| % Hit:                                | 100.00% | 100 | - |
| bin fired<br>Covered                  | 3       | 1   | - |
| Coverpoint cp_rx_ovf_irq<br>Covered   | 100.00% | 100 | - |
| covered/total bins:                   | 1       | 1   | - |
| missing/total bins:                   | 0       | 1   | - |
| % Hit:                                | 100.00% | 100 | - |
| bin fired<br>Covered                  | 5       | 1   | - |
| Coverpoint cp_done_irq<br>Covered     | 100.00% | 100 | - |
| covered/total bins:                   | 1       | 1   | - |
| missing/total bins:                   | 0       | 1   | - |



|                                                                             |         |     |   |
|-----------------------------------------------------------------------------|---------|-----|---|
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 36      | 1   | - |
| Covered                                                                     |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_interrupt_sources | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 5       | 5   | - |
| missing/total bins:                                                         | 0       | 5   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_irq                                                  | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 36      | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_rx_full_irq                                                   | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 4       | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_tx_ovf_irq                                                    | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |
| bin fired                                                                   | 3       | 1   | - |
| Covered                                                                     |         |     |   |
| Coverpoint cp_rx_ovf_irq                                                    | 100.00% | 100 | - |
| Covered                                                                     |         |     |   |
| covered/total bins:                                                         | 1       | 1   | - |
| missing/total bins:                                                         | 0       | 1   | - |
| % Hit:                                                                      | 100.00% | 100 | - |

|                                                  |         |     |   |
|--------------------------------------------------|---------|-----|---|
| bin fired                                        | 5       | 1   | - |
| Covered                                          |         |     |   |
| Coverpoint cp_done_irq                           | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| bin fired                                        | 36      | 1   | - |
| Covered                                          |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_irq_w1c | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 5       | 5   | - |
| missing/total bins:                              | 0       | 5   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_w1c                       | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_rx_full_w1c                        | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_tx_ovf_w1c                         | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |
| % Hit:                                           | 100.00% | 100 | - |
| Coverpoint cp_rx_ovf_w1c                         | 100.00% | 100 | - |
| Covered                                          |         |     |   |
| covered/total bins:                              | 1       | 1   | - |
| missing/total bins:                              | 0       | 1   | - |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_done_w1c                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_irq_w1c | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 5       | 5   | - |
| missing/total bins:                                               | 0       | 5   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_w1c                                        | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| Coverpoint cp_rx_full_w1c                                         | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 1       | 1   | - |
| Covered                                                           |         |     |   |
| Coverpoint cp_tx_ovf_w1c                                          | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 1       | 1   | - |
| missing/total bins:                                               | 0       | 1   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin cleared                                                       | 2       | 1   | - |
| Covered                                                           |         |     |   |

|                                                     |         |     |   |
|-----------------------------------------------------|---------|-----|---|
| Coverpoint cp_rx_ovf_w1c                            | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 1       | 1   | - |
| missing/total bins:                                 | 0       | 1   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin cleared                                         | 1       | 1   | - |
| Covered                                             |         |     |   |
| Coverpoint cp_done_w1c                              | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 1       | 1   | - |
| missing/total bins:                                 | 0       | 1   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin cleared                                         | 1       | 1   | - |
| Covered                                             |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_irq_masked |         |     |   |
|                                                     | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 10      | 10  | - |
| missing/total bins:                                 | 0       | 10  | - |
| % Hit:                                              | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_masked                       | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 2       | 2   | - |
| missing/total bins:                                 | 0       | 2   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin fired_unmasked                                  | 26      | 1   | - |
| Covered                                             |         |     |   |
| bin fired_masked                                    | 10      | 1   | - |
| Covered                                             |         |     |   |
| Coverpoint cp_rx_full_masked                        | 100.00% | 100 | - |
| Covered                                             |         |     |   |
| covered/total bins:                                 | 2       | 2   | - |
| missing/total bins:                                 | 0       | 2   | - |
| % Hit:                                              | 100.00% | 100 | - |
| bin fired_unmasked                                  | 2       | 1   | - |

|                                                                      |         |     |   |
|----------------------------------------------------------------------|---------|-----|---|
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 2       | 1   | - |
| Covered                                                              |         |     |   |
| Coverpoint cp_tx_ovf_masked                                          | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 2       | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 1       | 1   | - |
| Covered                                                              |         |     |   |
| Coverpoint cp_rx_ovf_masked                                          | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 2       | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 3       | 1   | - |
| Covered                                                              |         |     |   |
| Coverpoint cp_done_masked                                            | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 2       | 2   | - |
| missing/total bins:                                                  | 0       | 2   | - |
| % Hit:                                                               | 100.00% | 100 | - |
| bin fired_unmasked                                                   | 28      | 1   | - |
| Covered                                                              |         |     |   |
| bin fired_masked                                                     | 8       | 1   | - |
| Covered                                                              |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_irq_masked | 100.00% | 100 | - |
| Covered                                                              |         |     |   |
| covered/total bins:                                                  | 10      | 10  | - |
| missing/total bins:                                                  | 0       | 10  | - |
| % Hit:                                                               | 100.00% | 100 | - |
| Coverpoint cp_tx_empty_masked                                        | 100.00% | 100 | - |
| Covered                                                              |         |     |   |

|                              |         |     |   |
|------------------------------|---------|-----|---|
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 26      | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 10      | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_rx_full_masked | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 2       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_tx_ovf_masked  | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 1       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_rx_ovf_masked  | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |
| missing/total bins:          | 0       | 2   | - |
| % Hit:                       | 100.00% | 100 | - |
| bin fired_unmasked           | 2       | 1   | - |
| Covered                      |         |     |   |
| bin fired_masked             | 3       | 1   | - |
| Covered                      |         |     |   |
| Coverpoint cp_done_masked    | 100.00% | 100 | - |
| Covered                      |         |     |   |
| covered/total bins:          | 2       | 2   | - |

|                                                                 |         |     |   |
|-----------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                             | 0       | 2   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| bin fired_unmasked                                              | 28      | 1   | - |
| Covered                                                         |         |     |   |
| bin fired_masked                                                | 8       | 1   | - |
| Covered                                                         |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_delay                  | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| Coverpoint cp_delay                                             | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| bin delay_zero                                                  | 47      | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_min                                                   | 4       | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_small                                                 | 6       | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_medium                                                | 52      | 1   | - |
| Covered                                                         |         |     |   |
| bin delay_large                                                 | 58      | 1   | - |
| Covered                                                         |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_delay | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |
| Coverpoint cp_delay                                             | 100.00% | 100 | - |
| Covered                                                         |         |     |   |
| covered/total bins:                                             | 5       | 5   | - |
| missing/total bins:                                             | 0       | 5   | - |
| % Hit:                                                          | 100.00% | 100 | - |

|                                                    |         |     |   |
|----------------------------------------------------|---------|-----|---|
| bin delay_zero                                     | 47      | 1   | - |
| Covered                                            |         |     |   |
| bin delay_min                                      | 4       | 1   | - |
| Covered                                            |         |     |   |
| bin delay_small                                    | 6       | 1   | - |
| Covered                                            |         |     |   |
| bin delay_medium                                   | 52      | 1   | - |
| Covered                                            |         |     |   |
| bin delay_large                                    | 58      | 1   | - |
| Covered                                            |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_ss_select | 100.00% | 100 | - |
| Covered                                            |         |     |   |
| covered/total bins:                                | 11      | 11  | - |
| missing/total bins:                                | 0       | 11  | - |
| % Hit:                                             | 100.00% | 100 | - |
| Coverpoint cp_ss_en                                | 100.00% | 100 | - |
| Covered                                            |         |     |   |
| covered/total bins:                                | 11      | 11  | - |
| missing/total bins:                                | 0       | 11  | - |
| % Hit:                                             | 100.00% | 100 | - |
| bin slave0_only                                    | 44      | 1   | - |
| Covered                                            |         |     |   |
| bin slave1_only                                    | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin slave2_only                                    | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin slave3_only                                    | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin no_slave                                       | 35      | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[3]                                 | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[5]                                 | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[6]                                 | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[9]                                 | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[10]                                | 1       | 1   | - |
| Covered                                            |         |     |   |
| bin multi_slave[12]                                | 2       | 1   | - |
| Covered                                            |         |     |   |



|                                                                     |         |     |   |
|---------------------------------------------------------------------|---------|-----|---|
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_ss_select | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 11      | 11  | - |
| missing/total bins:                                                 | 0       | 11  | - |
| % Hit:                                                              | 100.00% | 100 | - |
| Coverpoint cp_ss_en                                                 | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 11      | 11  | - |
| missing/total bins:                                                 | 0       | 11  | - |
| % Hit:                                                              | 100.00% | 100 | - |
| bin slave0_only                                                     | 44      | 1   | - |
| Covered                                                             |         |     |   |
| bin slave1_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin slave2_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin slave3_only                                                     | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin no_slave                                                        | 35      | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[3]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[5]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[6]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[9]                                                  | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[10]                                                 | 1       | 1   | - |
| Covered                                                             |         |     |   |
| bin multi_slave[12]                                                 | 2       | 1   | - |
| Covered                                                             |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_loopback                   | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 11      | 11  | - |
| missing/total bins:                                                 | 0       | 11  | - |
| % Hit:                                                              | 100.00% | 100 | - |
| Coverpoint cp_loopback                                              | 100.00% | 100 | - |
| Covered                                                             |         |     |   |
| covered/total bins:                                                 | 2       | 2   | - |

|                                                                    |         |     |   |
|--------------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                                | 0       | 2   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| bin off                                                            | 7       | 1   | - |
| Covered                                                            |         |     |   |
| bin on                                                             | 38      | 1   | - |
| Covered                                                            |         |     |   |
| Coverpoint cp_width                                                | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 3       | 3   | - |
| missing/total bins:                                                | 0       | 3   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| bin w8                                                             | 23      | 1   | - |
| Covered                                                            |         |     |   |
| bin w16                                                            | 11      | 1   | - |
| Covered                                                            |         |     |   |
| bin w32                                                            | 11      | 1   | - |
| Covered                                                            |         |     |   |
| Cross cx_loopback_width                                            | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 6       | 6   | - |
| missing/total bins:                                                | 0       | 6   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| Auto, Default and User Defined Bins:                               |         |     |   |
| bin <on,w32>                                                       | 9       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w32>                                                      | 2       | 1   | - |
| Covered                                                            |         |     |   |
| bin <on,w16>                                                       | 9       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w16>                                                      | 2       | 1   | - |
| Covered                                                            |         |     |   |
| bin <off,w8>                                                       | 3       | 1   | - |
| Covered                                                            |         |     |   |
| bin <on,w8>                                                        | 20      | 1   | - |
| Covered                                                            |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_loopback | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 11      | 11  | - |
| missing/total bins:                                                | 0       | 11  | - |

|                                      |         |     |   |
|--------------------------------------|---------|-----|---|
| % Hit:                               | 100.00% | 100 | - |
| Coverpoint cp_loopback               | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 2       | 2   | - |
| missing/total bins:                  | 0       | 2   | - |
| % Hit:                               | 100.00% | 100 | - |
| bin off                              | 7       | 1   | - |
| Covered                              |         |     |   |
| bin on                               | 38      | 1   | - |
| Covered                              |         |     |   |
| Coverpoint cp_width                  | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 3       | 3   | - |
| missing/total bins:                  | 0       | 3   | - |
| % Hit:                               | 100.00% | 100 | - |
| bin w8                               | 23      | 1   | - |
| Covered                              |         |     |   |
| bin w16                              | 11      | 1   | - |
| Covered                              |         |     |   |
| bin w32                              | 11      | 1   | - |
| Covered                              |         |     |   |
| Cross cx_loopback_width              | 100.00% | 100 | - |
| Covered                              |         |     |   |
| covered/total bins:                  | 6       | 6   | - |
| missing/total bins:                  | 0       | 6   | - |
| % Hit:                               | 100.00% | 100 | - |
| Auto, Default and User Defined Bins: |         |     |   |
| bin <on,w32>                         | 9       | 1   | - |
| Covered                              |         |     |   |
| bin <off,w32>                        | 2       | 1   | - |
| Covered                              |         |     |   |
| bin <on,w16>                         | 9       | 1   | - |
| Covered                              |         |     |   |
| bin <off,w16>                        | 2       | 1   | - |
| Covered                              |         |     |   |
| bin <off,w8>                         | 3       | 1   | - |
| Covered                              |         |     |   |
| bin <on,w8>                          | 20      | 1   | - |
| Covered                              |         |     |   |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_tx_fifo                  | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_tx_fifo | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_tx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 8       | 8   | - |
| missing/total bins:                                               | 0       | 8   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin tx_empty                                                      | 40      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_one_word                                                   | 40      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_partial                                                    | 88      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_mid                                                        | 22      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_almost_full                                                | 11      | 1   | - |
| Covered                                                           |         |     |   |
| bin tx_full                                                       | 7       | 1   | - |
| Covered                                                           |         |     |   |
| bin empty_to_full                                                 | 1       | 1   | - |
| Covered                                                           |         |     |   |
| bin full_to_empty                                                 | 5       | 1   | - |
| Covered                                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_rx_fifo                  | 100.00% | 100 | - |

|                                                                   |         |     |   |
|-------------------------------------------------------------------|---------|-----|---|
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_rx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_rx_fifo | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| Coverpoint cp_rx_count                                            | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 7       | 7   | - |
| missing/total bins:                                               | 0       | 7   | - |
| % Hit:                                                            | 100.00% | 100 | - |
| bin rx_empty                                                      | 40      | 1   | - |
| Covered                                                           |         |     |   |
| bin rx_one_word                                                   | 13      | 1   | - |
| Covered                                                           |         |     |   |
| bin rx_partial                                                    | 17      | 1   | - |
| Covered                                                           |         |     |   |
| bin rx_almost_full                                                | 6       | 1   | - |
| Covered                                                           |         |     |   |
| bin rx_full                                                       | 6       | 1   | - |
| Covered                                                           |         |     |   |
| bin empty_to_full                                                 | 5       | 1   | - |
| Covered                                                           |         |     |   |
| bin full_to_empty                                                 | 5       | 1   | - |
| Covered                                                           |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_busy                     | 100.00% | 100 | - |
| Covered                                                           |         |     |   |
| covered/total bins:                                               | 2       | 2   | - |

|                                                                |         |     |   |
|----------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| Coverpoint cp_busy                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| bin idle                                                       | 200     | 1   | - |
| Covered                                                        |         |     |   |
| bin active                                                     | 137     | 1   | - |
| Covered                                                        |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_busy | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| Coverpoint cp_busy                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 2       | 2   | - |
| missing/total bins:                                            | 0       | 2   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| bin idle                                                       | 200     | 1   | - |
| Covered                                                        |         |     |   |
| bin active                                                     | 137     | 1   | - |
| Covered                                                        |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_transfer_done         | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 1       | 1   | - |
| missing/total bins:                                            | 0       | 1   | - |
| % Hit:                                                         | 100.00% | 100 | - |
| Coverpoint cp_done                                             | 100.00% | 100 | - |
| Covered                                                        |         |     |   |
| covered/total bins:                                            | 1       | 1   | - |

|                                                                         |         |     |   |
|-------------------------------------------------------------------------|---------|-----|---|
| missing/total bins:                                                     | 0       | 1   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| bin done                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_transfer_done | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 1       | 1   | - |
| missing/total bins:                                                     | 0       | 1   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| Coverpoint cp_done                                                      | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 1       | 1   | - |
| missing/total bins:                                                     | 0       | 1   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| bin done                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reg_written                    | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 7       | 7   | - |
| missing/total bins:                                                     | 0       | 7   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| Coverpoint cp_reg_written                                               | 100.00% | 100 | - |
| Covered                                                                 |         |     |   |
| covered/total bins:                                                     | 7       | 7   | - |
| missing/total bins:                                                     | 0       | 7   | - |
| % Hit:                                                                  | 100.00% | 100 | - |
| bin ctrl                                                                | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin tx_data                                                             | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin clk_div                                                             | 43      | 1   | - |
| Covered                                                                 |         |     |   |
| bin ss_ctrl                                                             | 85      | 1   | - |
| Covered                                                                 |         |     |   |

|                                                                       |         |     |   |
|-----------------------------------------------------------------------|---------|-----|---|
| bin int_en                                                            | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin int_stat                                                          | 1       | 1   | - |
| Covered                                                               |         |     |   |
| bin delay_reg                                                         | 43      | 1   | - |
| Covered                                                               |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reg_written |         |     |   |
|                                                                       | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 7       | 7   | - |
| missing/total bins:                                                   | 0       | 7   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| Coverpoint cp_reg_written                                             | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 7       | 7   | - |
| missing/total bins:                                                   | 0       | 7   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| bin ctrl                                                              | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin tx_data                                                           | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin clk_div                                                           | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin ss_ctrl                                                           | 85      | 1   | - |
| Covered                                                               |         |     |   |
| bin int_en                                                            | 43      | 1   | - |
| Covered                                                               |         |     |   |
| bin int_stat                                                          | 1       | 1   | - |
| Covered                                                               |         |     |   |
| bin delay_reg                                                         | 43      | 1   | - |
| Covered                                                               |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reg_read                     | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 8       | 8   | - |
| missing/total bins:                                                   | 0       | 8   | - |
| % Hit:                                                                | 100.00% | 100 | - |
| Coverpoint cp_reg_read                                                | 100.00% | 100 | - |
| Covered                                                               |         |     |   |
| covered/total bins:                                                   | 8       | 8   | - |
| missing/total bins:                                                   | 0       | 8   | - |



|                                                                    |         |     |   |
|--------------------------------------------------------------------|---------|-----|---|
| % Hit:                                                             | 100.00% | 100 | - |
| bin ctrl                                                           | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin status                                                         | 40      | 1   | - |
| Covered                                                            |         |     |   |
| bin rx_data                                                        | 48      | 1   | - |
| Covered                                                            |         |     |   |
| bin clk_div                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin ss_ctrl                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_en                                                         | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_stat                                                       | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin delay_reg                                                      | 6       | 1   | - |
| Covered                                                            |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reg_read |         |     |   |
|                                                                    | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 8       | 8   | - |
| missing/total bins:                                                | 0       | 8   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| Coverpoint cp_reg_read                                             | 100.00% | 100 | - |
| Covered                                                            |         |     |   |
| covered/total bins:                                                | 8       | 8   | - |
| missing/total bins:                                                | 0       | 8   | - |
| % Hit:                                                             | 100.00% | 100 | - |
| bin ctrl                                                           | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin status                                                         | 40      | 1   | - |
| Covered                                                            |         |     |   |
| bin rx_data                                                        | 48      | 1   | - |
| Covered                                                            |         |     |   |
| bin clk_div                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin ss_ctrl                                                        | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_en                                                         | 6       | 1   | - |
| Covered                                                            |         |     |   |
| bin int_stat                                                       | 6       | 1   | - |
| Covered                                                            |         |     |   |

|                                                                        |         |     |   |
|------------------------------------------------------------------------|---------|-----|---|
| bin delay_reg                                                          | 6       | 1   | - |
| Covered                                                                |         |     |   |
| TYPE /tb_top_sv_unit/spi_coverage_col/cg_reset_values                  | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| Coverpoint cp_reset_read                                               | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| bin ctrl_reset                                                         | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin status_reset                                                       | 40      | 1   | - |
| Covered                                                                |         |     |   |
| bin clkdiv_reset                                                       | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin ssctrl_reset                                                       | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin inten_reset                                                        | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin intstat_reset                                                      | 6       | 1   | - |
| Covered                                                                |         |     |   |
| bin delay_reset                                                        | 6       | 1   | - |
| Covered                                                                |         |     |   |
| Covergroup instance \tb_top_sv_unit::spi_coverage_col::cg_reset_values | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |
| Coverpoint cp_reset_read                                               | 100.00% | 100 | - |
| Covered                                                                |         |     |   |
| covered/total bins:                                                    | 7       | 7   | - |
| missing/total bins:                                                    | 0       | 7   | - |
| % Hit:                                                                 | 100.00% | 100 | - |

|                   |    |   |   |
|-------------------|----|---|---|
| bin ctrl_reset    | 6  | 1 | - |
| Covered           |    |   |   |
| bin status_reset  | 40 | 1 | - |
| Covered           |    |   |   |
| bin clkdiv_reset  | 6  | 1 | - |
| Covered           |    |   |   |
| bin ssctrl_reset  | 6  | 1 | - |
| Covered           |    |   |   |
| bin inten_reset   | 6  | 1 | - |
| Covered           |    |   |   |
| bin intstat_reset | 6  | 1 | - |
| Covered           |    |   |   |
| bin delay_reset   | 6  | 1 | - |
| Covered           |    |   |   |

TOTAL COVERGROUP COVERAGE: 100.00% COVERGROUP TYPES: 15

#### ASSERTION RESULTS:

| Name                                                                      | File(Line)                | Failure     | Pass       | Vacuous |
|---------------------------------------------------------------------------|---------------------------|-------------|------------|---------|
| Disable                                                                   | Attempt                   | Active Peak | Active ATV |         |
| Count                                                                     | Count                     | Count       | Count      | Count   |
| -----                                                                     |                           |             |            |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_1_SETUP_THEN_ACCESS      |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(32) | 0           | 3129374    | 6270705 |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_2_PENABLE_NEEDS_PSEL     |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(40) | 0           | 3129375    | 6270704 |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_3a_ADDR_CTRL_STABLE      |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(48) | 0           | 3129374    | 6270705 |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_3b_WDATA_STABLE_ON_WRITE |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(56) | 0           | 2119       | 9397960 |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_4a_PRDATA_RESERVED_ZERO  |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(65) | 0           | 5          | 9400074 |
| 70                                                                        | 9400149                   | 0           | 1 off      |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_4b_PRDATA_WO_REG_ZERO    |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(74) | 0           | 4          | 9400075 |
| 70                                                                        | 9400149                   | 0           | 1 off      |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_5_PREADY_ALWAYS_1        |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(82) | 0           | 3129375    | 6270704 |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/APB_6_PSLVERR_ALWAYS_0       |                           |             |            |         |
|                                                                           | assertions/spi_sva.sv(90) | 0           | 9400079    | 0       |
| 70                                                                        | 9400149                   | 0           | 10 off     |         |

|                                                                              |         |                            |        |         |         |  |
|------------------------------------------------------------------------------|---------|----------------------------|--------|---------|---------|--|
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_1_TX_OVF_SET_ON_OVERFLOW    |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(98)  | 0      | 3       | 9400076 |  |
| 70                                                                           | 9400149 | 0                          | 2 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_2_IRQ_EQUATION              |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(106) | 0      | 9400079 | 0       |  |
| 70                                                                           | 9400149 | 0                          | 10 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3a_TX_EMPTY_STICKY          |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(115) | 0      | 9395463 | 4614    |  |
| 70                                                                           | 9400149 | 2                          | 18 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3b_RX_FULL_STICKY           |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(124) | 0      | 1410    | 9398669 |  |
| 70                                                                           | 9400149 | 0                          | 4 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3c_TX_OVF_STICKY            |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(133) | 0      | 225     | 9399854 |  |
| 70                                                                           | 9400149 | 0                          | 4 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3d_RX_OVF_STICKY            |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(142) | 0      | 1562    | 9398517 |  |
| 70                                                                           | 9400149 | 0                          | 2 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_3e_DONE_STICKY              |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(151) | 0      | 8859498 | 540579  |  |
| 70                                                                           | 9400149 | 2                          | 18 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/IRQ_4_INT_STAT_ZERO_AFTER_RESET |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(159) | 0      | 10      | 9400069 |  |
| 70                                                                           | 9400149 | 0                          | 10 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/FIFO_2_RX_COUNT_BOUNDED         |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(167) | 0      | 9400079 | 0       |  |
| 70                                                                           | 9400149 | 0                          | 10 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_regfile/u_regfile_sva/FIFO_3_RX_OVF_SET_ON_OVERFLOW   |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(175) | 0      | 24      | 9400055 |  |
| 70                                                                           | 9400149 | 0                          | 1 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_1_SCLK_IDLE_POLARITY              |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(202) | 0      | 8327    | 9391752 |  |
| 70                                                                           | 9400149 | 0                          | 10 off |         |         |  |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_2A_MOSI_STABLE_RISING_SAMPLE      |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(210) | 0      | 5144    | 9394935 |  |
| 70                                                                           | 9400149 | 0                          | 9 off  |         |         |  |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_2B_MOSI_STABLE_FALLING_SAMPLE     |         |                            |        |         |         |  |
|                                                                              |         | assertions/spi_sva.sv(218) | 0      | 224     | 9399855 |  |

|                                                                         |         |   |                            |   |     |         |
|-------------------------------------------------------------------------|---------|---|----------------------------|---|-----|---------|
| 70                                                                      | 9400149 | 0 | 1 off                      |   |     |         |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_3_START_REQUIRES_SS_ASSERTED |         |   |                            |   |     |         |
|                                                                         |         |   | assertions/spi_sva.sv(226) | 0 | 621 | 9399458 |
| 70                                                                      | 9400149 | 0 | 9 off                      |   |     |         |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_5_TX_POP_ONE_CYCLE_PULSE     |         |   |                            |   |     |         |
|                                                                         |         |   | assertions/spi_sva.sv(234) | 0 | 621 | 9399458 |
| 70                                                                      | 9400149 | 0 | 9 off                      |   |     |         |
| /tb_top/u_wrap/u_dut/u_core/u_core_sva/SPI_6_DONE_PULSE_ONE_CYCLE       |         |   |                            |   |     |         |
|                                                                         |         |   | assertions/spi_sva.sv(242) | 0 | 621 | 9399458 |
| 70                                                                      | 9400149 | 0 | 9 off                      |   |     |         |

Total Coverage By Instance (filtered view): 93.11%

End time: 17:18:13 on May 21,2026, Elapsed time: 0:00:00

Errors: 0, Warnings: 0

===== GRADER COVERAGE SUMMARY =====

Functional Coverage: 100.00%

Total Coverage: 93.11%

Code Coverage: See DUT statement/branch coverage sections above.

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